



**parkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA
University of Pangasinan**

A Project Presented to the
Department of Computer Engineering
College of Engineering and Architecture
University of Pangasinan

In Partial Fulfillment of
the Requirements for the Degree
Bachelor of Science in Computer Engineering

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ACKNOWLEDGEMENT

The researchers express their sincere gratitude to all who contributed to the completion of this study, entitled ParkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan.

We extend our heartfelt thanks to our adviser, Engr. Yves Lyndon P. Abalos, for his guidance, support, and invaluable insights throughout the research process.

We also acknowledge the faculty and staff of the College of Engineering and Architecture (CEA), as well as the Information Technology Services Department (ITSD) and General Service Department (GSD), for their resources, assistance, and support.

Our deepest appreciation goes to our families, classmates, and friends for their encouragement and understanding.

Lastly, we recognize the dedication and cooperation of each group member, whose collective effort made this study possible.



DEDICATION

This project is dedicated to our families, whose unwavering support, understanding, and encouragement made this endeavor possible. Their constant belief in our abilities inspired us to persevere and complete this study successfully.

We also dedicate this work to our instructors and mentors, whose knowledge, guidance, and patience greatly contributed to the development of this research. Their teachings have helped shape our skills and prepared us for future professional challenges.

Lastly, this study is dedicated to the students, educators, and school administrators. May this system serve as a step toward a more secure, efficient, and accessible academic environment. We hope that parkhAlve will help address modern educational challenges with practical, innovative, and cost-effective solutions.



ABSTRACT

Title: parkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan

Keywords: Machine Learning, Image Processing, Digital Twin, Parking Management System, Computer Vision

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This study presents parkhAIve, a smart parking management system developed to address parking congestion, inefficient space utilization, and limited real-time visibility within PHINMA University of Pangasinan. The system utilizes image processing and machine learning technologies to enable real-time slot detection and vehicle monitoring in a campus-based parking environment. parkhAIve employs a 1080p IP camera and a YOLOv8n-based object detection model, running on a local processing server, to accurately detect vehicles, determine parking slot occupancy, and assess slot usability. QR code-based vehicle validation features were incorporated to enhance access control and user interaction. A web application-based digital twin interface was developed to visualize live parking conditions, manage user data, and support administrative operations.

System evaluation was conducted through functional, accuracy, reliability, and maintainability testing, along with a user survey involving 44 respondents: 40 students, 2 security guards, and 2 ITS employees. Results show that parkhAIve achieved 80% detection accuracy and maintained real-time performance, with an average latency of 15 seconds, and received positive usability ratings from participants. The findings demonstrate effective hardware-to-software integration and improved parking efficiency.

The study concludes that parkhAIve is a cost-effective, scalable, and practical smart parking solution that supports campus mobility and contributes to digital transformation initiatives.



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CHAPTER I

INTRODUCTION

Background of the Study

In the bustling maze of modern cities, finding a parking space has become a daily battle that numerous drivers spend time, money, and patience on. According to Market Reports World, more than 1.4 billion registered vehicles were operating worldwide in 2023, creating daily parking challenges in densely populated cities. While cities struggle with parking challenges, university campuses also face similar crises, including limited parking, rising vehicle numbers, and growing frustration among students and faculty. 60% of college students believe their campus lacks adequate parking, with many forced to park off-campus or arrive late due to traffic congestion (Ezarik et al., 2022). These inefficiencies have a worldwide impact, making smart parking solutions more than a convenience.

Traditional manual parking management is labor-intensive, error-prone, and provides little real-time visibility. Manual systems struggle to scale with fluctuating demand during class changes, exam periods, and events. Incorporating machine learning improves the accuracy of vehicle detection and tracking and enables automatic detection of parking slot availability. According to Knights, Petrovska, Bunevska-Talevska, and Prchkovska (2024), integrating machine learning with IoT can significantly improve parking space prediction and management. Their study, *Machine Learning Models and*



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Mathematical Approaches for Predictive IoT Smart Parking, compared several models and found that the LightGBM algorithm achieved the highest accuracy in forecasting parking availability. The researchers concluded that combining ML with IoT reduces congestion and enhances the efficiency of smart parking systems.

The prototype development of *parkhAIve*, a smart parking management system, aims to provide real-time parking operations tailored to PHINMA Avenue at PHINMA University of Pangasinan. Parking inefficiencies in academic institutions often lead to congestion, wasted time, and user frustration. *parkhAIve*'s specifications allow the university to streamline slot detections. By offering a scalable and cost-effective platform, *parkhAIve* will leverage image processing, machine learning, and a web application-based digital twin interface to enhance parking operations, reduce traffic impacts, and promote digital transformation within the campus community.



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OBJECTIVES OF THE STUDY

General Objective

To design and develop a prototype smart parking management system that employs image processing and machine learning for real-time monitoring of parking slot availability. The system synchronizes real-time detection with a web application-based digital twin interface to provide updated parking slot information and support parking management.

Specific Objective

1. To design and develop a prototype smart parking management system using YOLOv8n for real-time vehicle detection and occupancy monitoring.
2. To implement a real-time data acquisition pipeline using a network-connected IP camera to support a continuous image stream for the computer vision model.
3. To train a YOLOv8n model to achieve improved detection accuracy in identifying four-wheeled private vehicles which will help in classifying parking slots as occupied or available.
4. To design and develop a web-based interface that displays real-time parking slot availability and system information for users and administrators.
5. To evaluate the system's performance in terms of detection accuracy, response time, and detection consistency.



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Significance of the Study

In the global context, the study is beneficial in addressing the challenges in urban mobility and resource efficiency. It aligns with the UN Sustainable Development Goals, particularly SDG 9: Industry, Innovation, and Infrastructure, by promoting smart and efficient parking solutions. Additionally, it contributes to sustainable cities, advances innovation and infrastructure, and supports climate action through reduced traffic congestion and vehicle emissions.

In the economic context, the study contributes to the development of cost-effective, scalable, and adaptable systems that improve operational efficiency and support the growth of smart city economies. By automating slot detection and real-time monitoring, the system reduces the need for manual recording of vehicle entries and exits. This allows universities to optimize labor allocation, minimize human error, and reduce operational costs while ensuring accurate and reliable parking management.

In the environmental context, the study aims to lessen air pollution and carbon emissions by reducing traffic congestion and unnecessary fuel consumption. In addition, efficient parking management promotes environmental protection, sustainable urban mobility, and resource conservation.



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In the societal context, the study contributes to enhancing urban liability, sustainability, and public well-being. The system reduces driver stress, saves time, and promotes safer, more organized communities through efficient, transparent parking management. In the long run, it improves the overall parking experience for everyone involved.

Scope and Delimitation

Scope

The study focuses on the prototype development of a smart parking management system deployed on a local processing server, capable of monitoring parking slot availability using an IP camera and a YOLOv8n-based object detection model, enhanced with multi-object tracking integrated with Re-Identification (ReID) for vehicle identification across frames. The system provides real-time parking slot detection and supports parking management by displaying available and occupied slots within the selected area.

The system integrates real-time parking slot monitoring using an IP camera that captures and transmits a live video feed to a functioning local processing server. The server processes the data using a YOLOv8n-based object detection model, while ByteTrack and ReID handle tracking and identity consistency, updating slot status throughout the system. OpenCV is utilized for video stream processing, frame handling,



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and detection support. This setup provides users with real-time parking information through a web application interface hosted on a locally deployed server and accessible for testing via a development tunneling service.

By supporting more efficient parking management, the system aims to assist in improving parking flow within the selected area. Furthermore, the study demonstrates the feasibility of applying machine learning-based infrastructure in localized institutional environments.

The system is developed as a base model for smart campus infrastructure. It establishes a foundational architecture for real-time synchronization between physical parking slots and the system interface, providing a scalable framework that may be extended to other parking zones within the institution or adapted for future intelligent environment applications.

Delimitation

The study is designed for a campus-based parking management system implemented within a designated parking area along PHINMA Avenue of PHINMA University of Pangasinan. The system's coverage is strictly limited to the parking area located along the University Gymnasium, measuring 22 meters in length and 5.5 meters in width, with a total area of 121 square meters.



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The system is implemented as a prototype in a controlled campus environment where user access and parking activities can be monitored. It is not intended for large-scale municipal, commercial, or public parking infrastructures. Furthermore, the system does not account for external variables such as off-campus traffic conditions, weather changes, or environmental factors that may affect parking behavior. It also assumes that users and administrators possess basic digital literacy.

The system relies on compliance with the prescribed process, such as scanning QR codes prior to parking. Non-compliance, including bypassing QR scanning or entering the parking area without initiating a session, is considered outside the system's automated tracking scope. A manual override mechanism is provided to handle such cases; however, accurate session tracking primarily depends on system-enforced procedures.

In terms of functionality, the system is limited to detecting four-wheel vehicles only. All other objects and vehicle types, including bicycles, pedestrians, signage, and environmental elements such as trees, are treated as non-vehicle objects and are not classified by the system. The system does not perform license plate recognition, vehicle identification, or vehicle-type classification beyond occupancy detection. Additionally, parallel parking configurations are not included within the detection and slot management logic of the system.



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The study is intended for testing and evaluation purposes only and does not represent a full production deployment. Access to the system is provided through a development tunneling service using Visual Studio Code port forwarding, allowing temporary access to the locally hosted server for testing and validation. The system is not hosted on a public production server and is not intended for continuous public use.

Lastly, the study is limited to specific hardware components due to resource constraints.

Conceptual Framework

INPUT	PROCESS	OUTPUT
Users - Security Guards - Students - Employees Software - MySQL - Node.js - React.js - Laravel - OpenCV - Roboflow - ByteTrack - ReID - YOLOv8n - HTTPSMS Server Hardware 1080p IP Camera - Local Processing	- Planning and designing the project - Gathering of materials and parts - Design and development of a software database system - Testing and configuring hardware components - Integrating hardware and software - Testing and debugging of software and hardware - Adjustments and	parkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan



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• Server • Router • Network Switch	• revisions • Final testing and evaluation of project software in real-world applications	
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Figure 1.1 Research Paradigm of the Study

This study develops the aforementioned system using a structured Input-Process-Output (IPO) approach. The Input phase consists of key users such as security guards, students, and employees, alongside essential software components such as MySQL for database management, Node.js for server-side operations, React.js for the digital twin interface, Laravel for core backend logic, Roboflow for dataset annotation, and artificial intelligence components, including the YOLOv8n model, ByteTrack, and Re-Identification (ReID). The YOLOv8n model is used for real-time vehicle detection in every video frame, with the main custom-trained model handling parking detection and a lightweight reference model assisting with missed-vehicle detection. ByteTrack is used for multi-object tracking, ensuring stable, consistent vehicle IDs across consecutive frames within the same camera, while a Re-Identification (ReID) module extracts appearance embeddings to maintain consistent vehicle identities within a single camera view, particularly when combined with QR-based vehicle tagging. The hardware components consist of a 1080p IP camera for vehicle monitoring, a local processing server for process control, a router for network management, an SMS server for automated SMS notifications, and a software database for data storage and retrieval. The



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Process phase covers project planning and design, gathering of required materials, development of the software database system, testing and configuration of hardware components, and integration of hardware and software modules, followed by debugging refinement, and validation to ensure reliability and accuracy, leading to final testing and real-world evaluation. The Output phase is the fully realized parkhAIve system, integrating intelligent detection, tracking, and identification mechanisms to deliver an efficient and reliable smart parking management solution for the university.

Definition of Terms

ByteTrack - A multi-object tracking algorithm that assigns consistent IDs to detected objects across video frames, enabling reliable tracking of vehicles over time in a vision-based parking management system.

Digital Twin - A real-time digital image of the PHINMA Avenue parking slots that shows which slots are occupied and which cars can fit in the vacant slots.

Local Processing Server - A desktop-based local server deployed on-site that performs real-time data processing and AI inference near the data source, enabling low-latency system response.

Image Processing - An algorithm used to analyze and interpret images, such as enhancing detection accuracy.



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Laravel - A framework used to develop the web application of the parking management system.

Machine Learning - A method used to train the system to make decisions, such as identifying parking slot occupancy and determining which cars can fit in the slot.

MySQL - A relational database used to store and manage parking data, enabling real-time updates between the system's components.

Node.js - A tool that helps build and prepare the React frontend.

Parking Management - The process of handling parking slot usage efficiently and effectively.

React - A JavaScript library used to develop the digital twin's frontend interface.

ReID (Re-Identification) - An AI technique that extracts vehicle appearance features to match vehicles across multiple frames, enabling consistent identification, particularly after QR tagging.

Roboflow - An online tool used for annotating datasets for machine learning.

YOLOv8n - A machine learning tool, which is a lightweight version of YOLOv8, used to train models for vehicle detection, car classification, and slot occupancy detection.



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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter is composed of relevant literature and studies that form the foundation of this study. Through a critical analysis of existing literature and studies, it identifies knowledge gaps, establishes theoretical connections, and highlights the study's significance.

Computer Vision

Computer Vision serves as the "eyes" of the parkhAIve system, providing the primary sensing capability for monitoring PHINMA Avenue. Unlike traditional sensor-based systems that require expensive hardware for each slot, computer vision uses existing or strategically placed IP cameras to monitor large areas simultaneously. By applying deep learning algorithms to live video feeds, the system can autonomously identify vehicle presence, track movements, and determine slot availability. This software-driven approach is not only more cost-effective but also more scalable, allowing the university to adapt the system to different parking layouts without invasive infrastructure changes.

In a study by Sobirin et al. (2024), the researchers conducted an extensive comparative analysis of different versions of the YOLO architecture, demonstrating that YOLOv8 significantly outperforms earlier models such as YOLOv5 in both precision and



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recall. Their study found that YOLOv8's enhanced feature extraction is particularly effective in complex parking environments where vehicles may be partially occluded. This supports the use of YOLOv8n in ParkhAIve, as it provides accurate detection while remaining computationally efficient, making it suitable for real-time parking slot monitoring on limited hardware.

The efficiency of "Nano" models was specifically addressed by Sani and Ayyasy (2024), who validated that YOLOv8n (Nano) is the optimal choice for edge computing devices. Their research showed that despite its small parameter count, the model maintains a high mean Average Precision (mAP) while requiring significantly less power and processing time. This supports parkhAIve's deployment on a local processing server, enabling high-speed inference without the need for expensive, high-end GPU clusters.

Popereshnyak et al. (2025) explored integrating YOLOv8 with the ByteTrack algorithm to maintain consistent vehicle identification across video frames. Their findings indicated that tracking-based systems are much more resilient to temporary obstructions, such as a pedestrian walking in front of a camera. This is critical for parkhAIve, as it ensures that a parked car remains "occupied" in the system even if the view is momentarily blocked.

In a move toward privacy-conscious monitoring, Bakirci (2024) demonstrated that Re-Identification (ReID) techniques can successfully track vehicles across non-overlapping campus cameras without using license plates. By extracting deep visual



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features like color, shape, and size, the system can "remember" a vehicle as it moves from one camera's view to another. This study provides the theoretical foundation for parkhAIve's ability to maintain vehicle IDs throughout the PHINMA Avenue gym area without compromising user privacy.

The scalability of wide-angle computer vision was highlighted by MDPI (2025), which utilized panoramic vision and YOLO-based models to monitor up to 50 parking slots from a single elevated camera. Their research showed that a bird's-eye-view approach reduces the number of hardware units needed for full coverage.

Acharya et al. (2023) focused on the importance of "Region of Interest" (RdiagramOI) mapping in computer vision for parking. By pre-defining the exact coordinates of each parking slot within the camera's field of view, the system can ignore irrelevant background movement and focus its processing power only on the slots. This optimization technique is a core part of parkhAIve's workflow, ensuring that the system is both accurate and computationally efficient.

Research from IEEE Xplore (2024) emphasized the benefits of edge analytics in smart campus environments. By processing video feeds locally rather than sending them to the cloud, the researchers avoided bandwidth congestion on the university's network and ensured lower latency. This study validates parkhAIve's architectural decision to use a local desktop as a local processing server, making the system more reliable in an institutional setting with limited internet bandwidth.



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Rizkatama et al. (2023) developed a system that allows users to interact with computer vision data using natural language queries. Their study showed that by connecting YOLO detections to a structured database, users could ask the system simple questions about slot availability. While parkhAIve uses a visual interface, this research underscores the versatility of computer vision data when stored in a modern database such as MySQL.

Chen and Chang (2023) addressed the challenge of "Temporal Continuity" in outdoor parking environments. They argued that for a system to be reliable, it must track a vehicle over time rather than just in individual frames. Their use of tracking algorithms ensures that environmental factors, such as heavy rain or shadows, do not cause the system to "lose" a vehicle. This supports ParkhAIve's implementation of continuous monitoring to provide a stable and accurate parking occupancy status.

Finally, a 2026 study on UAV-based vision systems demonstrated that modern YOLO models are now capable of "Small Object Detection" from significant distances. This indicates that even if the IP cameras are mounted high on the gym walls, the system will maintain high accuracy in detecting vehicles. This research provides confidence in the hardware placement strategies for parkhAIve, ensuring that the system can handle the spatial constraints of the campus environment.

The use of Computer Vision is the technological cornerstone of this study, providing a non-intrusive, cost-effective, and highly accurate method for managing



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campus parking by leveraging state-of-the-art models like YOLOv8n and ByteTrack. parkhAIve moves beyond the limitations of manual recording and expensive sensors. This keyword justifies the system's ability to provide real-time, automated monitoring that is both scalable for the university and respectful of the community's privacy.

Digital Twin

In a study by Chornobryvets et al. (2025), the researchers developed a dynamic digital twin framework for urban parking that focused on real-time occupancy synchronization. Their system allowed administrators to not only view current slot status but also run “what-if” simulations to predict how traffic would flow during high-demand events. This study highlights the transition of parking management from static reporting to active, predictive modeling, which is essential for managing the fluctuating traffic of a university campus.

A smart campus initiative documented by MDPI (2025) at Chiang Mai University successfully implemented a modular digital twin to enhance facility management and campus mobility. The researchers found that providing a virtual replica of campus assets reduced the time managers spent on physical site inspections by 30%. The study serves as a benchmark for how localized digital twins can significantly improve the operational efficiency of university security personnel.



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Zhang and Wu (2024) explored a specialized four-layer digital twin architecture consisting of the Physical, Data, Model, and Service layers. Their research specifically addressed the challenges of underground and complex parking environments, emphasizing that the accuracy of the twin depends on the high-fidelity mapping between physical sensors and virtual entities. Their findings support the use of structured data pipelines to ensure the digital twin remains a reliable source of truth for the end-user.

The technical feasibility of web-based digital twins was further examined by Li et al. (2025), who focused on the use of React.js for building “thin-client” dashboards. They argued that web-based platforms offer superior cross-device accessibility for campus communities compared to resource-heavy standalone software. This study justifies the selection of React.js for the parkhAIve interface, ensuring that students and employees can access the parking twin from any mobile or desktop browser.

Addressing the predictive capabilities of virtual modeling, the Urban Mobility Lab (2025) utilized digital twins to forecast parking demand during peak university hours. By feeding historical occupancy data into the virtual model, the system could alert drivers of potential congestion before they even arrived on campus. This emphasizes the value of a digital twin not just for present monitoring, but as a tool for proactive traffic management.

Guo et al. (2023) introduced the concept of "privacy-by-design" within digital twin frameworks, suggesting that raw video feeds should be replaced by abstracted 2D or



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38 icons. This approach ensures that while vehicle presence and slot occupancy are tracked, individual privacy is maintained. This study is highly relevant to parkhAIve's goal of providing a secure monitoring environment that does not rely on sensitive personal identifiers or license plate data.

In the context of spatial navigation, the Smart City Report (2024) validated that integrating real-time data with a 3D Building Information Model (BIM) allows for enhanced "spatial awareness." This allows drivers to navigate to a specific empty slot with precision rather than circling a general zone. Their research highlights how digital twins can drastically reduce the "search time" that contributes to campus-wide traffic congestion.

Vidyavani et al. (2023) conducted a critical analysis of synchronization latency in digital twin systems, concluding that edge-processed data reduces the delay between a car parking and the dashboard update to under 2 seconds. This research supports parkhAIve's use of a local processing server to ensure that the digital twin provides near-instantaneous feedback to users, which is vital for a responsive parking experience.

The environmental impact of digital twin implementation was studied by Gonzalez et al. (2024) in a campus setting in Madrid. Their findings revealed that real-time vacancy mapping leads to a measurable reduction in carbon emissions by minimizing the "cruising" time of vehicles. This study strengthens the environmental



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significance of parkhAIve, aligning it with broader sustainability goals through optimized urban mobility.

Lastly, a study published in Preprints.org (2025) demonstrated a cross-platform twinning mechanism that connects computer vision inferences directly to a database for automated facility reports. This research proves that the integration of AI and digital twinning can automate administrative tasks that were previously manual. This supports the parkhAIve objective of reducing the workforce required for routine parking record-keeping and audit logs.

The adoption of Digital Twin technology is indispensable for parkhAIve as it transforms abstract data into a functional user experience. By synthesizing real-time detection with a virtual interface, the system provides the "visual intelligence" necessary to manage PHINMA University of Pangasinan's parking resources effectively. Without this keyword, the system would remain a backend tool; with it, it becomes an accessible, transparent, and efficient platform for the entire campus community.

Image Processing

Image processing, the use of computers to manipulate digital images for enhancement or information extraction, has become an integral technology in modern parking management systems. Tagolanao et al. (2021) demonstrated its application in



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their study *Development of an Automatic Data-Centered Parking Space Occupancy Detection with Vehicle License Plate Web-Based Monitoring*. Their system integrates a retrained YOLO (You Only Look Once) model with Tesseract OCR (Optical Character Recognition) to detect license plates from video feeds and extract character information. These details are logged in a database along with entry and exit timestamps. Additionally, the system employs a simplified AlexNet, a type of Convolutional Neural Network (CNN), to classify parking slots as “vacant” or “occupied,” enabling real-time monitoring of space availability displayed on driver-facing monitors.

Expanding on this concept, Prasetyo et al. (2021), in *Optimization of Image Processing Techniques in Developing Smart Parking Systems*, sought to improve both efficiency and security in parking management. They addressed vulnerabilities in manual and camera-assisted systems, such as ticket fraud and license plate manipulation, by combining OCR with SURF (Speeded Up Robust Features). OCR automatically reads and records license plate numbers, while SURF captures unique physical characteristics of vehicles, even those of identical make and color. This dual-algorithm approach enhances system security by supplementing existing hardware and software with robust image-based verification.

Other researchers have explored new ways of refining image-based detection accuracy. He, Liu, and Wang (2024), in *A Parking Space Recognition Method Based on Digital Image Technology*, proposed a parking space recognition system divided into key



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processes: pre-processing, parking space division, CNN model construction and training, and parking space recognition. Since the dataset used was small, the authors applied data augmentation to enhance model accuracy. Similarly, Zhang et al. (2023) developed an image processing algorithm for parking lot detection that improves image equalization through filtering and binarization, emphasizing optimization techniques for more reliable slot recognition.

The evolution of parking management reflects a clear progression from hardware-intensive, sensor-based methods to software-driven, vision-based approaches. Early methods, including induction loops, ultrasonic detectors, and infrared or microwave sensors, offered accurate vehicle detection but required invasive installation, were prone to environmental interference, and incurred high maintenance costs (Chiu et al., Park et al., Boda et al., as cited in Sree Chandra et al., 2025). Wireless sensor networks improved scalability but still lacked slot-level precision and real-time availability. Vision-based systems emerged as cost-effective alternatives, using cameras and image processing to identify vacant spaces. However, their performance has remained sensitive to shadows, occlusions, and lighting variability (Zhang et al., as cited in Sree Chandra et al., 2025). To overcome these issues, enhancements such as clustering algorithms, region of interest (ROI) models, unsupervised learning, visual markers, and reference image matching have been proposed (Fabian, Hafizi, Yusnita et al., as cited in Sree Chandra et al., 2025). Parallel developments in license plate recognition (LPR) have integrated hybrid fuzzy



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logic, template matching, and neural networks, though challenges persist with non-standard plates and adverse conditions (Sree Chandra et al., 2025).

Recent innovations highlight the adaptability of software-driven solutions. B. Kavya Sri et al. (2025) proposed an image-processing framework using OpenCV in Python combined with coordinate-bound pixel analysis, edge detection, and Tesseract OCR to detect occupancy and extract textual data from images. Their system leverages variable-level image processing to adapt to different environmental conditions, enhancing scalability across diverse parking layouts from open lots to multi-story garages.

Suhana et al. (2025) developed an IoT-based emergency vehicle detection system using the YOLOv8 deep learning algorithm, achieving 95% detection accuracy at 31 frames per second. Their architecture, which employed a Pi Camera and Raspberry Pi for edge processing, demonstrates the feasibility of deploying lightweight, locally available hardware for real-time vehicle recognition. This approach is highly relevant to *ParkhAlve*, which aims to integrate YOLOv8n for automated entry and exit monitoring within PHINMA University of Pangasinan's campus. By adapting their edge-based setup, *ParkhAlve* can implement cost-effective vehicle detection modules that support blockchain-logged access control and priority-based slot allocation. Furthermore, the system's validation through confusion matrix analysis and performance metrics provides a benchmark for evaluating *ParkhAlve's* own computer vision component, ensuring operational accuracy and responsiveness in a smart campus environment.



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Talha et al. (2025) introduced an automated parking management system that leverages deep learning for real-time image processing and OCR to detect vacant spaces and recognize vehicle number plates. Their system achieved a 95.6% accuracy in space detection and 92.3% in plate recognition, demonstrating its robustness in both simulated and real-world environments. The integration of Bengali and English plate formats highlights its adaptability across diverse urban contexts.

In a similar study, Rani et al. (2024) integrated deep learning with image processing to advance the efficiency of object detection in parking systems. For instance, the YOLO (You Only Look Once) algorithm leverages image processing to analyze entire images at once, generating bounding boxes and class probabilities with high speed and accuracy. This approach eliminates the need for region-by-region analysis, making it computationally efficient and ideal for real-time applications (Rani et al., 2024). By refining raw images through preprocessing techniques such as noise reduction, grayscale conversion, and resizing, image processing ensures higher reliability in detection tasks across diverse conditions.

Amarasooriya and Peiris (2023) proposed a cost-effective and accurate smart parking system leveraging image processing techniques. Their study utilized video footage as input and applied the YOLO v3 object detection algorithm to identify parking slots and vehicles. By comparing the coordinates of detected vehicles and predefined parking spaces, the system could determine slot availability in real time. The model was



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trained using the PKLot dataset, which includes images under various weather conditions, and achieved an average accuracy of 88.01%, with optimal performance on sunny days and reduced accuracy during rainy conditions. This highlights the robustness and limitations of image-based systems under environmental variability. The integration of convolutional neural networks (CNNs) and real-time video analysis in this context demonstrates the evolving role of image processing in urban infrastructure. Unlike sensor-based systems, image processing offers scalability and reduced hardware dependency, making it suitable for large-scale implementations. Moreover, the use of YOLO v3—a real-time object detection algorithm—underscores the importance of speed and precision in dynamic environments like parking lots.

The integration of image processing technologies into smart parking systems has gained significant interest in recent years, offering scalable and efficient solutions to urban mobility challenges. Cruz et al. (2024) developed a vehicle parking monitoring and management system utilizing computer vision, specifically leveraging OpenCV and YOLOv8 for real-time vehicle detection and tracking. Their approach exemplifies the practical application of image processing in optimizing campus parking operations, reducing carbon emissions, and enhancing user accessibility. Cruz et al.'s (2024) system stands out by integrating image processing with a web-based interface, enabling real-time monitoring and historical data analytics. The use of YOLOv8, a state-of-the-art object detection model, demonstrates the evolution of image processing from traditional



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methods to AI-enhanced frameworks, offering both precision and scalability in parking management.

Machine Learning

The rapid increase in vehicle ownership, particularly in urban areas, has intensified the demand for efficient parking management systems capable of reducing congestion, minimizing search time, and optimizing space utilization. Traditional parking methods, which often lack real-time monitoring, contribute to traffic build-up as drivers circulate in search of available spaces, underscoring the need for smarter, technology-driven solutions.

One such advancement is the Smart Parking Management System (SPMS) developed by Cunanan et al. (2025). This system integrates camera-based monitoring with machine learning algorithms to detect vacant and occupied slots in real time. Its Parking Spot Detection (PSD) algorithm uses masked images and bounding box segmentation to classify spaces, while an Object and User Classifier (OUC) algorithm, powered by a YOLOv8 pretrained model, identifies vehicle types and categorizes users such as teaching staff, non-teaching staff, and students. Evaluation through ISO 25010 quality metrics revealed “Excellent” ratings across functional suitability, performance efficiency, reliability, and usability, reflecting both high operational effectiveness and strong user acceptance. By combining automated visual detection with user classification,



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the SPMS offers a scalable, accurate, and user-friendly approach to campus parking management, aligning with smart city initiatives.

The research conducted by Shankar et al. (2024) presents an improved car parking detection system that utilizes the deep learning capabilities of YOLOv8. The model was trained using a diverse set of annotated parking lot images to accurately identify vehicles and determine whether parking slots were occupied or vacant. By applying YOLOv8's real-time object detection, the system achieved high precision in recognizing vehicle presence under various lighting conditions and camera angles. This study demonstrates how properly trained deep learning models can significantly enhance the accuracy and responsiveness of smart parking systems, making them suitable for dynamic environments such as university campuses.

Similarly, Balaji et al. (2025) emphasized predictive management by proposing a machine learning-driven optimization framework for large institutions. Their system integrates IoT sensing with supervised learning algorithms to analyze both real-time and historical usage data, incorporating contextual factors such as time of day, day of the week, and location-specific demand. The framework forecasts occupancy levels across campus zones and provides pre-arrival guidance for drivers through mobile applications. This predictive approach reduces search time, alleviates congestion, improves space utilization, and lowers carbon emissions. Moreover, the system continuously learns from



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new data, enhancing accuracy and scalability, making it highly adaptive to evolving traffic patterns in smart campus environments.

Beyond predictive models, several studies highlight the effectiveness of computer vision for parking detection. Dasari et al. (2023) introduced a Python and OpenCV–based Smart Parking System that employed libraries such as Numpy, CVZone, and Pickle to enable accurate readings, handle diverse data types, and support Python object serialization. In a similar effort, Srushti et al. (2024) developed a vision-based detection method using grayscale conversion to simplify image data, Gaussian blurring to reduce noise, and thresholding to ensure reliable classification of parking availability.

The study by Al Mudawi et al. (2023) presents a robust approach to vehicle detection and classification using YOLOv8 combined with a Deep Belief Network (DBN), applied to aerial image sequences. YOLOv8 is used to detect vehicles in real time, while the DBN handles classification based on features extracted through SIFT, ORB, and KAZE descriptors. The system was trained on a diverse dataset and optimized to perform accurately under varying environmental conditions. This layered architecture demonstrates the effectiveness of combining object detection and feature-based classification to enhance traffic monitoring and vehicle analysis, which can be adapted for smart parking systems that require precise identification of vehicle types and occupancy status.



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Bakirci (2024) evaluated YOLOv8n, the smallest and fastest variant of the YOLOv8 family, for real-time vehicle detection in Intelligent Transportation Systems. Despite its compact architecture, YOLOv8n achieved over 80% detection accuracy, demonstrating its viability for resource-constrained environments where speed and efficiency are critical. The study emphasized YOLOv8n's suitability for real-time applications through parametric analysis of image processing time and detection sensitivity.

Tomas et al. (2024) explored the application of YOLOv8 for vehicle type detection using dashcam footage captured in highway and evening settings. The study demonstrated YOLOv8's high detection accuracy—exceeding 90% for individual vehicle categories—by leveraging its one-stage architecture, Feature Pyramid Network (FPN), and anchor box enhancements. The dataset was processed using Roboflow, enabling robust training and testing across varied environmental conditions. Although the research was conducted in a traffic surveillance context, its findings affirm YOLOv8's adaptability to ground-level implementations such as smart campus parking systems. For ParkhAIve, this supports the model's integration for real-time vehicle identification and classification in Philippine university settings.

Zhao et al. (2025) introduced YOLOv8-MAH, an enhanced deep learning model designed for multi-attribute vehicle recognition, addressing challenges in intra-class variability and inter-class similarity. By integrating MobileViT modules into the



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backbone and incorporating Efficient Multi-scale Attention (EMA) mechanisms, the model significantly improved its ability to extract fine-grained features such as vehicle brand, color, and direction. The addition of a specialized detection layer further refined attribute recognition, achieving strong performance across 144 categories in a custom dataset. Although developed for broader intelligent transportation applications, YOLOv8-MAH's architecture offers promising utility for smart campus parking systems like ParkhAIve, where detailed vehicle profiling supports access control, auditability, and enhanced user experience.

Neamah and Karim (2024) developed a real-time traffic monitoring system using YOLOv8's single-stage architecture to detect, classify, and count vehicles across four categories: sedan cars, buses, trucks, and motorcycles. The system achieved high performance, with detection accuracy at 96.58%, counting accuracy at 97.54%, and classification accuracy ranging from 94.7% to 99.7% depending on vehicle type. Deployed on an Nvidia GTX 1070 GPU, the model demonstrated practical feasibility for mid-range hardware setups. These results affirm YOLOv8's robustness in ground-level traffic environments, supporting its integration into smart campus parking systems like ParkhAIve for real-time vehicle monitoring, access control, and audit logging.

Machine learning emerges as the central component uniting these innovations, driving real-time monitoring, automated detection, and adaptive forecasting. Core algorithms such as PSD continuously classify parking spaces as either "empty" or



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“occupied” using masked images and bounding boxes, while visual indicators—green for available and red for occupied—improve usability. The OUC algorithm further enhances these systems by detecting vehicles and categorizing users, generating granular data to support more effective allocation policies. Together, these studies demonstrate how the integration of computer vision, IoT, and machine learning can transform parking management into an intelligent, efficient, and user-centered system capable of addressing the complex challenges of modern urban mobility.

Parking Management System

Research on Parking Management Systems (PMS) has increasingly emphasized the integration of emerging technologies to enhance efficiency, adaptability, and user experience. Wang et al. (2025) reviewed advancements in parking space detection, highlighting innovations that improve environmental adaptability and detection accuracy. Their findings indicate that modern PMS are evolving beyond simple monitoring to systems capable of adjusting to diverse urban conditions.

Expanding on this trend, Pradhan (2025) proposed a PMS framework that integrates IoT and image processing technologies with database systems to minimize inefficiencies and unauthorized usage. By leveraging automation, the system reduces the need for manual oversight while enabling real-time monitoring of parking slot availability.



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Similarly, Al Mamun et al. (2024) developed a smart PMS that connects sensors with mobile applications to support real-time monitoring and reservation. Their study demonstrates the practicality of IoT-enabled solutions in improving driver convenience and optimizing slot utilization, although they also highlight deployment costs as a barrier to large-scale implementation.

In line with this, Elsonbaty et al. (2020) designed an IoT-based Smart Parking Management System that allows users to locate vacant spaces through an Android mobile application. The system employs Arduino components and IR sensors to detect available slots, while a Wi-Fi module transmits data to a server for user access. This study underscores the role of low-cost hardware and mobile integration in enabling accessible parking solutions.

Building on web-based approaches, Luciano et al. (2023) introduced PARADAJuan, a parking lot management system developed using multi-paradigm programming languages such as HTML, CSS, JavaScript, PHP, and MySQL. Designed with the Agile Software Development Life Cycle (SDLC) and evaluated using ISO/IEC 25010 standards, PARADAJuan achieved high ratings in usability, functionality, and efficiency. Its browser-accessible interface supports real-time slot monitoring, reservation, and administrative control, demonstrating modularity and responsiveness. While it relies on relational databases and procedural logic, its framework provides a foundation for smart campus solutions. For instance, parkhAIve can build on



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PARADAJuan by integrating one-time passcode (OTP) authentication and computer vision to enhance transparency and automation within Philippine campus environments.

Rajendran et al. (2022) introduced a secure smart parking system built around a web application interface, enabling users to reserve slots, authenticate access, and retrieve parking data through a centralized portal. The system emphasizes secure data handling, user authentication, and modular zone management, ensuring both usability and integrity across multiple parking areas. While the paper focuses on the web layer, it implies backend coordination with IoT devices for real-time slot monitoring and gate control. This architecture aligns with parkhAIve's goals by illustrating how a secure web interface can enhance centralized system control, where administrators oversee real-time parking operations, enforce authentication, and maintain system integrity in campus-scale environments.

Venkata Sudhakar et al. (2023) developed a smart parking system that integrates a mobile-responsive web application with IoT-enabled hardware to streamline parking operations and enhance user experience. The web interface allows users to view real-time slot availability, pre-book parking spaces, receive safety alerts (e.g., fire or gas leakage), and access automated billing based on entry and exit timestamps. This front-end system is supported by a Raspberry Pi-based backend that processes sensor inputs and image data, transmitting relevant information to the cloud for remote access. Although the article does not specify the exact web technologies used, the architecture implies the use



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of standard web development stacks such as HTML, CSS, JavaScript, and cloud-based APIs. This modular and user-centric design aligns with the parkhAIve's goals by illustrating how a secure and scalable web interface can enable centralized system control, where administrators oversee operations while the YOLOv8n model facilitates efficient vehicle detection and real-time parking monitoring in campus environments.

Rafique et al. (2023) introduced a scalable smart parking framework that replaces traditional sensor-based systems with a deep learning-driven architecture centered on YOLOv5s. The system leverages existing CCTV infrastructure to detect vehicles in real time, achieving 96.8% accuracy and 45 FPS in live deployments. A custom post-processing algorithm maps detected vehicles to specific parking slots, enabling accurate occupancy tracking without physical sensors. The technology stack includes PyTorch for model deployment, OpenCV for image preprocessing, and a Python-based backend—likely Flask or Django—to serve a browser-accessible dashboard built with HTML, CSS, and JavaScript. This modular design supports real-time visualization and data flow via RESTful APIs, demonstrating how deep learning outputs can be translated into actionable web interfaces. This framework aligns with parkhAIve's goals by demonstrating the effectiveness of integrating the YOLOv8n model for real-time vehicle detection with a local processing server and a Laravel-based backend to ensure secure data handling and centralized system control in campus-scale deployments.



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Ali (2024) developed a vision-based smart parking system centered on a web application that integrates real-time image processing with user-facing functionality. The system utilizes surveillance camera feeds processed through OpenCV to detect vehicle presence and determine slot occupancy. This visual data is transmitted to a browser-accessible dashboard, allowing users to monitor parking availability in real time. The web application is built using standard technologies such as HTML, CSS, and JavaScript for the front-end, while the back-end likely employs Python-based frameworks (e.g., Flask or Django) to handle image data and API communication. The system architecture demonstrates how lightweight, camera-driven parking solutions can be deployed without extensive sensor infrastructure, offering a scalable and cost-effective alternative for campus environments.

Finally, Kumar and Sanketh (2021) developed an Online Parking Management System using a web-based software stack composed of HTML, CSS, JavaScript, jQuery, PHP, and MySQL, deployed locally via XAMPP. The system incorporates modules for administrators, users, and attendants, offering real-time slot booking, transaction tracking, and parking history management. Unified Modeling Language (UML) and Data Flow Diagrams (DFD) were used to model system interactions, ensuring clarity and maintainability. Their work reflects a conventional yet effective approach for parking automation in urban contexts.



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SYNTHESIS OF THE RELATED LITERATURE AND STUDIES

The present study builds on previous research emphasizing the use of advanced technologies to improve parking management systems in campus environments. Computer vision, particularly YOLOv8 and YOLOv8n, has been widely recognized for its ability to detect and track vehicles in real time, maintain temporal continuity, and optimize coverage with fewer cameras (Sobirin et al., 2024; Sani & Ayyasy, 2024; Popereshnyak et al., 2025). Privacy-preserving Re-identification techniques further allow vehicle monitoring without compromising user data (Bakirci, 2024). These studies highlight the cost-effectiveness, scalability, and reliability of vision-based approaches compared to traditional sensor-dependent methods.

Digital twin technology complements these systems by converting real-time detection into user-accessible interfaces, enabling predictive modeling, spatial awareness, and proactive parking management (Chornobryvets et al., 2025; MDPI, 2025; Li et al., 2025). Image processing and machine learning enhance detection accuracy and adaptability to environmental conditions, supporting real-time slot recognition, user classification, and predictive occupancy management (Tagolanao et al., 2021; Cunanan et al., 2025; Shankar et al., 2024; Balaji et al., 2025).

Parking Management Systems (PMS) integrating web-based and IoT interfaces demonstrate the importance of real-time monitoring, secure access, and user-friendly dashboards (Rajendran et al., 2022; Venkata Sudhakar et al., 2023; Kumar & Sanketh,



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2021). The present study combines these approaches by integrating computer vision, digital twin visualization, image processing, and machine learning into a comprehensive system for PHINMA University of Pangasinan. Unlike previous research that focuses on individual components, parkhAIve provides a fully automated, privacy-conscious, and scalable solution for real-time campus parking management, enhancing efficiency, accessibility, and sustainability.



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CHAPTER III

METHODOLOGY

This chapter comprehensively outlines the methodology employed in the study, encompassing the project's design materials and procedural framework. It further elaborates on the research and development process, project design, operational flow, material specifications, and the approaches utilized for project development, testing, and evaluation.

RESEARCH METHOD USED

The Research and Development (R&D) method is a structured approach to creating, refining, and validating new products, models, or services through systematic investigation. Commonly applied in fields such as education, technology, and applied sciences, it aims to address specific problems and produce solutions that are valid, practical, and effective for their intended application (Sugiyono, 2022).

This section presents the processes implemented by the researchers in developing *parkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan*.

The project followed a series of stages to achieve its objectives. These activities were carried out to enhance the system and guide its overall development. The researchers employed the Research and Development (R&D) method to identify effective



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solutions to the project's core issues. The phases involved in the R&D process—namely design, fabrication, and evaluation—were thoroughly discussed and implemented throughout the development of the system.

DESIGN PHASE

The design phase of *parkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan* focuses on defining the architecture, components, and interactions necessary to establish a fully functional, and user-friendly smart parking solution. This phase provides the blueprint for hardware and software integration, data management, and system operation, ensuring that the system fulfills its objectives. The system architecture comprises three major layers: hardware, software, and data management.

The hardware layer centers on the local processing server, which serves as the primary processing unit, handling image input, performing real-time analysis, and communicating with the backend system. The local processing server executes all core system functions, including vehicle detection and tracking, parking slot detection, and database operations, enabling centralized and low-latency processing.

Continuous video streams are captured along the designated monitoring zone on PHINMA Avenue and processed by the local processing server to support real-time vehicle monitoring and parking availability assessment. The processed data is



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synchronized with the backend system to update the dashboard and reflect current parking conditions.

The 1080p IP camera is installed at the Student Plaza, strategically positioned along the designated monitoring zone on PHINMA Avenue to capture incoming vehicles and monitor parking slot occupancy. This single camera serves a dual purpose: vehicle tagging and parking slot detection. However, its coverage is limited to the side of the University Gymnasium and does not capture the opposite side of the avenue. Meanwhile, QR code scanning is performed separately using the security guard's mobile device camera. This component enables the system to record time-in data in the database and link each vehicle to its corresponding digital session. Unregistered users, including visitors who do not work or study at PHINMA UPang, may be temporarily registered by the administrator with a single-session QR code, and their activity is monitored throughout their stay.

The featured 1080p IP camera is connected via gigabit ethernet to a network switch, which serves as the central data distribution hub. A router manages the IP address allocation and network traffic for connected devices.

The local processing server utilizes the YOLOv8n object detection model to identify vehicles, analyze parking slot occupancy, and determine slot availability. The system uses predefined parking slots to determine occupancy. Each slot is digitally mapped in the system, and a slot is marked as occupied once a vehicle covers most of the



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designated area, even if the vehicle is parked diagonally or slightly outside the boundaries. All processed data is then transmitted to the backend system for storage, synchronization, and visualization.

The software layer integrates multiple technologies to enable accurate vehicle detection, parking slot monitoring, and user interaction. A YOLO-based object detection pipeline, implemented using the Ultralytics YOLO framework, performs real-time vehicle detection from IP camera video streams to enable accurate parking slot occupancy analysis. To maintain consistent vehicle identities across consecutive frames within the camera, ByteTrack is employed for multi-object tracking, while a Re-Identification (ReID) module extracts vehicle appearance embeddings to assist in cross-camera identity matching and to support QR-based vehicle tagging. A web-based application serves as the primary user interface and features an interactive digital twin of the PHINMA Avenue parking area, specifically covering only the side adjacent to the University Gymnasium. It reflects live parking slot availability, occupancy status, and accessibility conditions based on real-time system updates. Rather than automatically assigning vehicles to specific slots, the system displays parkable zones based on current availability and detected conditions. Registered users can generate unique QR codes on the platform for entry logging and identity association, while non-registered users can view the digital twin but are limited to basic functionality.



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The data management layer ensures secure, structured, and reliable handling of system information. Vehicle detections, parking slot statuses, timestamps, QR-based entry records, and user interactions are stored in a MySQL database, which serves as the system's primary data repository. This centralized database architecture enables efficient data organization, retrieval, and updating, supporting real-time system operations and historical data analysis.

To maintain data integrity and consistency, structured schemas and relational constraints are implemented, ensuring accurate associations between vehicles, users, and parking sessions. The system also incorporates validation mechanisms to prevent erroneous or duplicate entries during data capture and processing. Additionally, access controls are enforced to prevent unauthorized access to the database, thereby enhancing data security and protecting sensitive user information.

The backend, implemented in Laravel with MySQL, manages system operations, data processing, and integration with AI modules such as YOLOv8n, ByteTrack, OpenCV, and Re-Identification (ReID) for real-time vehicle detection, tracking, image processing, and parking slot updates. The frontend, developed in React, provides a dynamic, responsive web interface that displays real-time parking availability and slot occupancy, allowing users and administrators to monitor vehicles and interact with the system efficiently.



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Overall, the design phase ensures that parkhAIve operates as a cohesive, intelligent, and adaptable smart parking management system. By combining multiple technologies, the system provides an innovative solution that reduces traffic congestion, minimizes search time for parking spaces, enhances user convenience, and improves overall parking efficiency within PHINMA University of Pangasinan.



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OPERATION FLOW

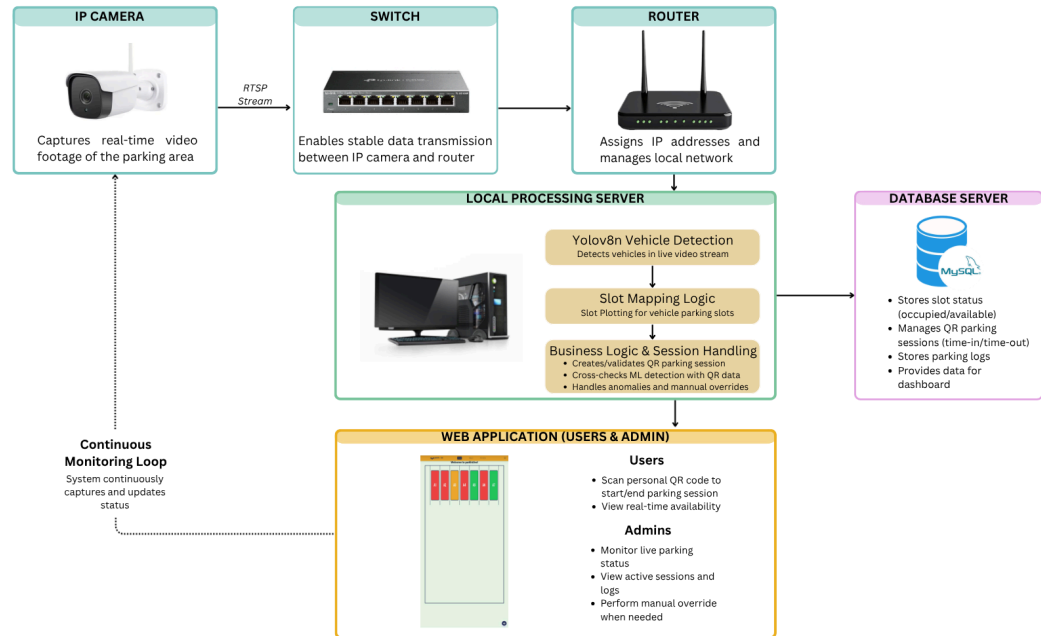


Figure 3.1 Block Diagram

The parkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan integrates multiple hardware components to enable automated parking management. At the university entrance, guards use mobile devices to scan user-generated QR codes via the web application.

The camera feed is transmitted through Gigabit Ethernet to the local processing server, which serves as the central processing unit. It runs the YOLOv8n model to detect vehicles and determine parking slot occupancy. The IP camera deployed along the



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designated monitoring zone on PHINMA Avenue provides continuous monitoring, improving detection accuracy even under partial occlusion or improper parking.

Processed data is reflected in the digital twin interface, displaying real-time occupancy. All system data, including logs and occupancy status, is stored in a centralized MySQL database via the backend server.



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Camera Flowchart:

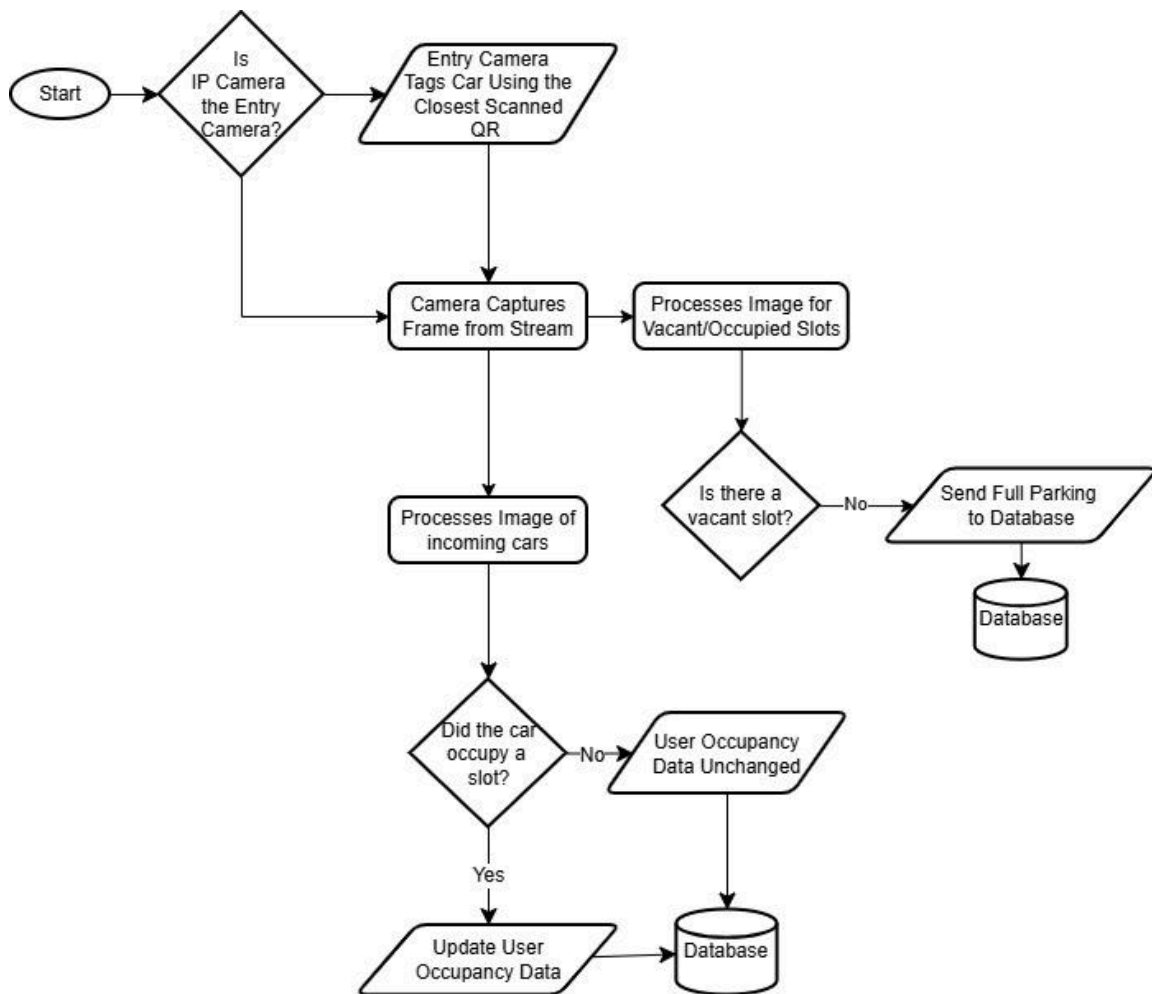


Figure 3.2 IP Camera Flowchart



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Software Interface Flowchart:

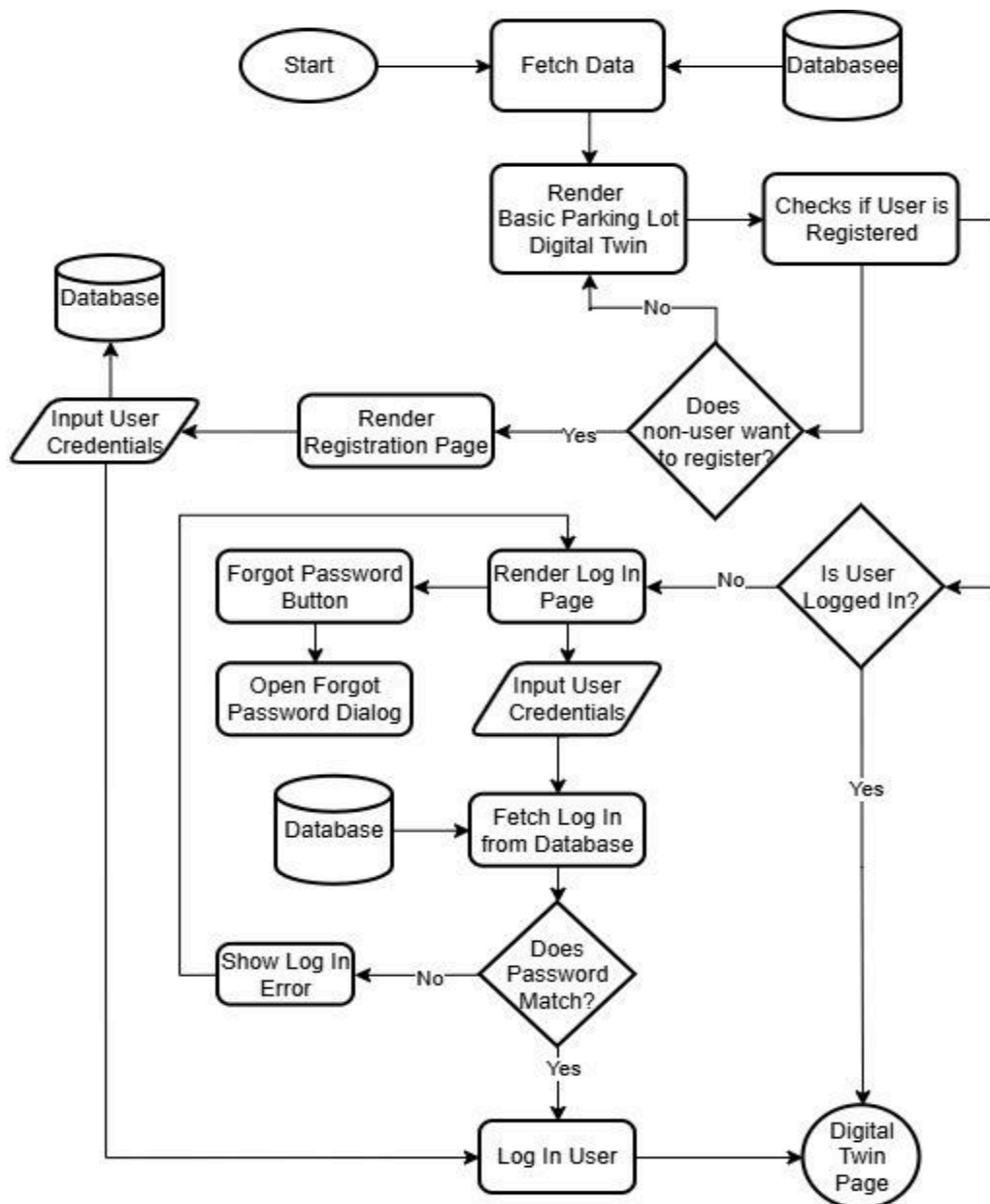


Figure 3.3 Software Interface Flowchart



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Digital Twin Interface Flowchart:

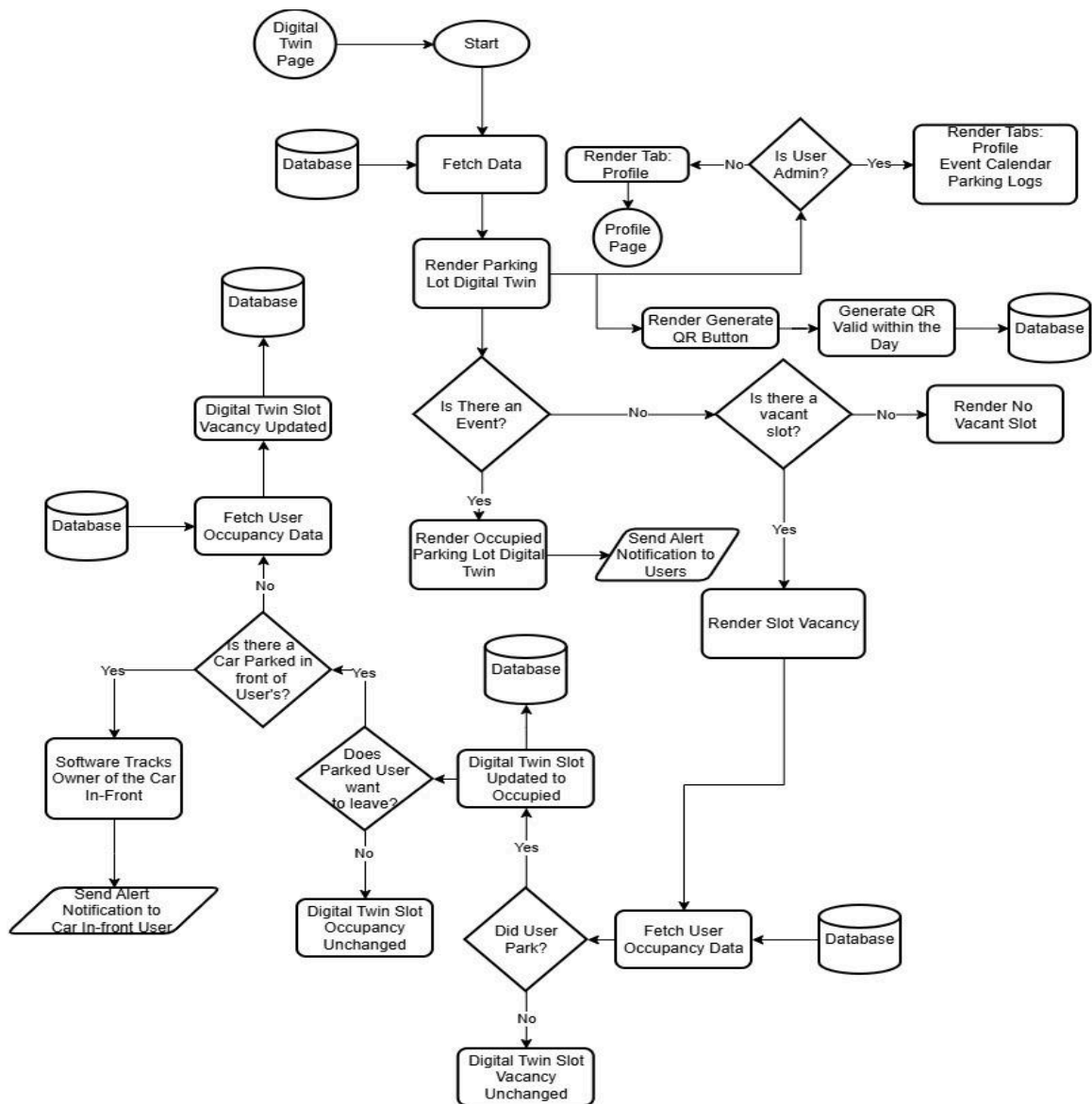


Figure 3.4 Digital Twin Interface Flowchart



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Admin Tabs Interface Flowchart:

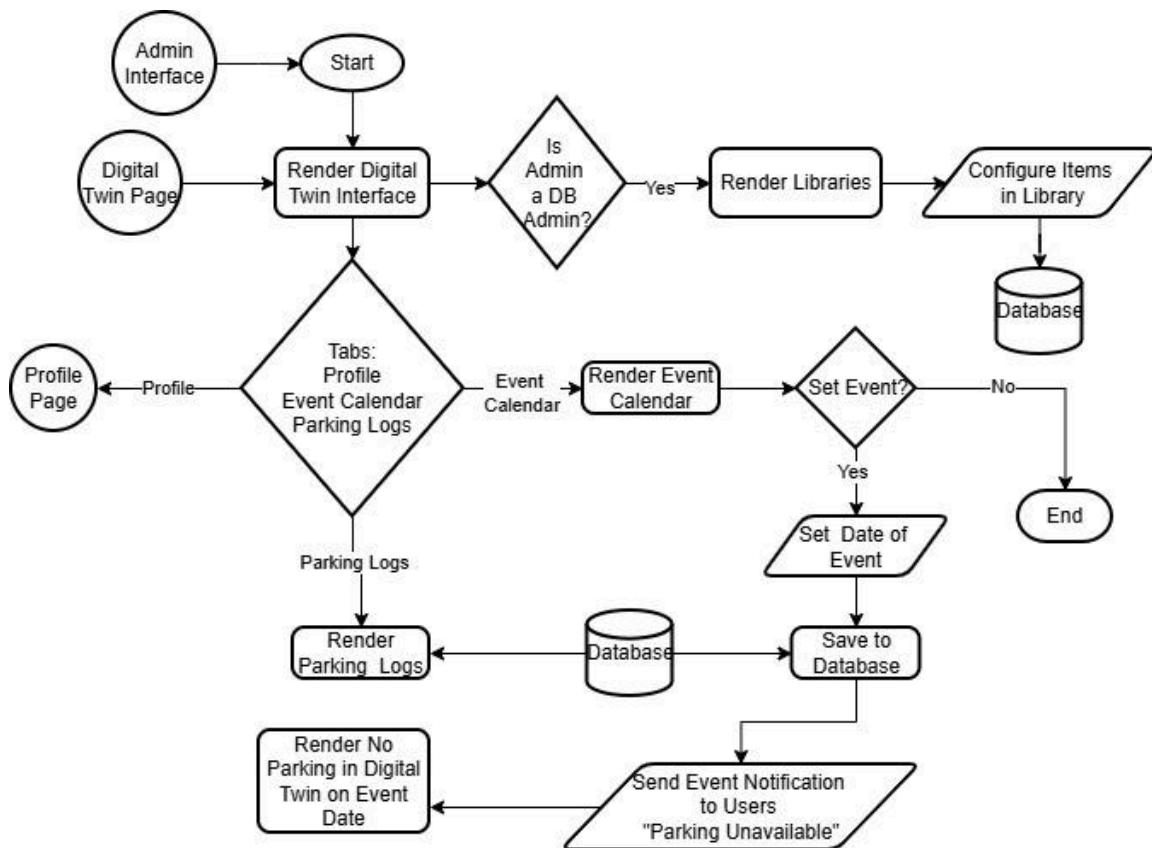


Figure 3.5 Admin Tabs Interface Flowchart



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Profile Page Flowchart:

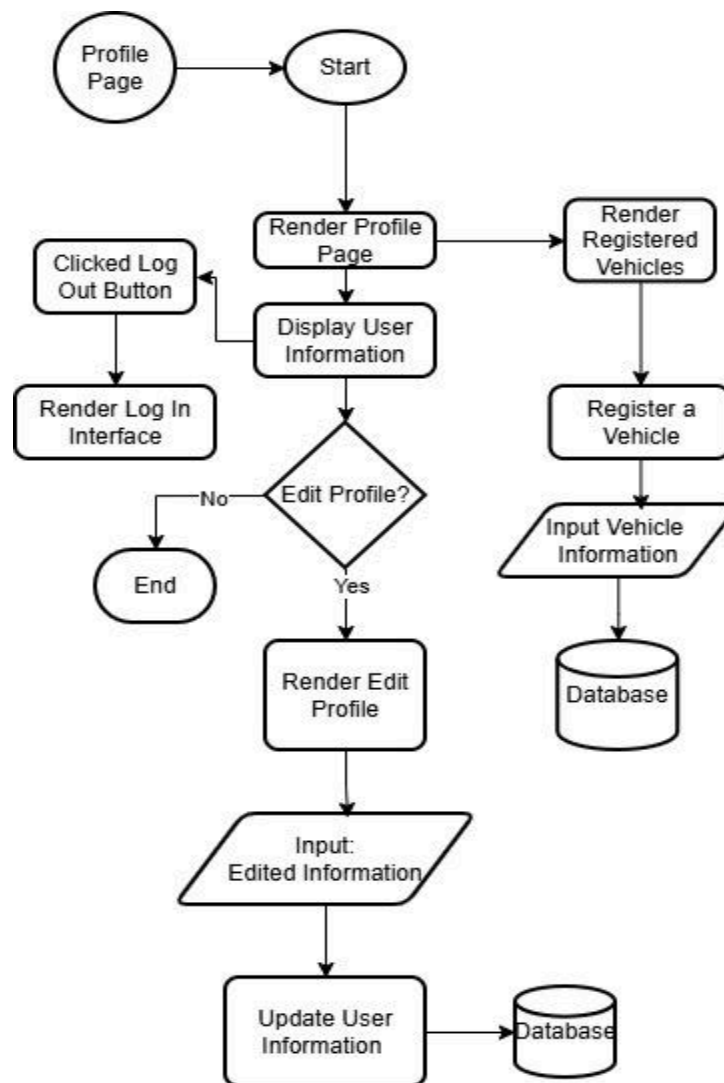


Figure 3.6 Profile Page Flowchart

ParkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan integrates QR code-based vehicle validation,



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real-time camera monitoring, and automated parking slot detection to provide an efficient and user-friendly parking management solution.

The system begins with user registration via the web application. Registered users can create an account, register multiple vehicles linked to unique license plates, and access full application features, including QR code generation, parking slot monitoring, and event notifications. Non-registered users can view the digital twin but cannot generate personal QR codes or receive event notifications.

Before entering the campus parking area, registered users generate a QR code valid only for the day of issuance. QR code generation is temporarily disabled when the parking area is full, and an in-page banner notifies users of this status. At the entrance, a guard scans the QR code using a mobile device, while the 1080p IP camera captures the vehicle in the detection zone. The system links the scanned QR code session to the detected vehicle, ensuring correct assignment even when similar-looking cars are present. For non-registered or occasional visitors, the admin collects vehicle and personal information to generate a temporary QR code valid for the duration of their stay.

Once validated, the vehicle is granted entry, and the digital twin begins tracking its position along the designated monitoring zone on PHINMA Avenue, specifically within the side adjacent to the University Gymnasium. The parking area in this section is defined using designated slot zones, which the system uses to detect and monitor occupancy. The 1080p IP camera continuously monitors the area, feeding live video to



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the local processing server, which runs YOLOv8n and image processing algorithms. The system detects private vehicles, evaluates slot usability, tracks movements between slots, and updates the digital twin in real time.

The system incorporates an event notification mechanism to alert both administrators and users of important parking-related events. For administrators, notifications appear alongside the digital twin interface, allowing continuous monitoring of parking occupancy and vehicle movement. For users, event notifications are displayed as a banner that temporarily overlays the digital twin interface to ensure critical updates are acknowledged. Additionally, SMS notifications are sent to users to provide off-platform alerts for improved accessibility and reliability.

To address real-world inconsistencies, the system includes a manual override feature. Authorized personnel can manually assign a registered user to a parking slot or terminate a parking session if the automated processes fail or are bypassed. This ensures continuity in parking management, accurate session tracking, and reliable data recording.

All parking activity data, including QR code entries, slot occupancy, entry times, vehicle movements, and notifications, is securely stored in a MySQL database.

By combining QR-based validation, camera verification, and adaptive slot tracking, the ParkhAIve system ensures accurate monitoring, optimal utilization of



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parking spaces, and a secure, efficient parking experience along the designated parking slot area on the University Gymnasium side of PHINMA Avenue.

DESCRIPTION AND SPECIFICATIONS OF MATERIALS USED

Table 3.1: List of Hardware Components Used

Electronic Component			
Component	Image	Description	Application
Local Processing Server		Consists of computer devices that serves as the central processing unit that runs the smart parking	It is used to detect parking slot occupancy, monitor vehicle entry, send processed data to a



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		software, manages databases, and processes machine learning based vehicle and slot data in real time.	central monitoring system, control system operations, and provide administrators with a live dashboard for efficient parking management.
1080p IP Camera		Captures images with a large number of pixels, resulting in greater detail and clarity. Measured in megapixels (MP).	Captures high-resolution video and images for vehicle detection and real-time monitoring
Gigabit Network Switch		A small device designed to connect an IP camera, a router, and a local processing server, enabling high-speed data transfer and reliable communication within the network.	Enables seamless communication between the IP camera, the router, and the local processing server. It supports real-time video transmission, centralized data processing, and reliable network connectivity.
CAT6 Ethernet Cable		A high-performance network cable designed for fast and reliable data transmission. Supports speeds of up to 1 Gbps.	Connects all hardware components of the ParkhAIve system, enabling communication and data transfer from the IP camera to the local processing



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			server.
Router		A network device that routes network connections between devices, establishing a static IP address and serving as the default gateway.	It is used to connect the networks of the local processing server, network switch, and IP camera

Table 3.1 shows the electronic components used in parkhAlve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan. It includes images of the components, descriptions, and the purpose and function of each piece of equipment in the system.



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Table 3.2: Bill of Materials

Bill of Materials			
Product Name	Unit Price	Quantity/Length	Total Price
PC Desktop	Php 28,000.00	1	Php 28,000.00
1080p IP Camera with SD Card (32GB)	Php 1,290.00	1	Php 1,290.00
TP-Link Litewave LS1008G Desktop Switch	Php 990.00	1	Php 990.00
Extension Wire (40 meters)	Php 25.00 per meters	40 meters	Php 1000.00
Extension Plug	Php 20.00	1	Php 20.00
Extension Outlet	Php 50.00	3	Php 150.00
CAT6 LAN Cable	Php 1,204.00	150 meters	Php 1,204.00
Total Cost of Materials			Php 32,654.00

Table 3.2 shows the cost of the materials used in developing the project, as well as the project's overall cost.



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FABRICATION PHASE

In developing parkhAIve: A YOLOv8n-Based Smart Parking Management System for PHINMA University of Pangasinan, the researchers first procured all the necessary hardware components and materials. After procurement, the researchers used die-cast cars for the initial testing of the system. This setup allowed the researchers to make quick adjustments and debug the system before conducting real-world testing. The researchers also created the network configuration setup for each hardware component. This allowed the researchers to begin initial testing of each hardware component, examine their interactions, and create a blueprint for LAN cable connections between components.

The camera was also tested to ensure that it could capture and accurately track vehicle data along the designated monitoring zone on PHINMA Avenue, specifically within the designated parking slot zones on the side adjacent to the University Gymnasium, while the web application was assessed for reliability and real-time performance.

By the end of the fabrication phase, all hardware components had been integrated and tested in preparation for the evaluation phase. During the evaluation phase, the entire



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system, including both software and hardware components, would be observed in a realistic, user-oriented environment.

Activities	August				September				October				November				December				January				February				March			
	Week1	Week2	Week3	Week4	Week1	Week2	Week3	Week4	Week1	Week2	Week3	Week4	Week1	Week2	Week3	Week4	Week1	Week2	Week3	Week4	Week1	Week2	Week3	Week4	Week1	Week2	Week3	Week4	Week1	Week2	Week3	Week4
Conducting of research																																
Title Presentation and Defense																																
Curvassing of Materials																																
Document Preparation																																
Submission and Clerkling of Documents																																
Developing of Prototype																																
Evaluation Process																																
Finalization of Documents																																
Document Presentation																																

Table 3.3: Thesis Project Gantt Chart

EVALUATION PHASE

During the evaluation phase, the integration and communication between the web application, hardware components, and backend system were thoroughly tested. Registered and non-registered users could view the real-time status of PHINMA Avenue through the digital twin interface when they opened the web application and accessed features such as QR generation, slot monitoring, and notifications.

A single IP camera positioned along the designated monitoring zone on PHINMA Avenue monitored the parking area and streamed live video to a local processing server,



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where the YOLOv8n model was used for real-time vehicle detection and parking slot analysis. Upon entry, users scanned their QR codes via a guard's mobile device, while the same 1080p IP camera verified and associated the scanned QR code with the detected vehicle in the scene.

For temporary users, the admin generated and managed their QR code. Their exits were manually recorded through the system dashboard, which logged their time-out, updated the digital twin, and invalidated their temporary record. The SMS server was also tested for reliability, ensuring that OTP and event notifications were properly sent to vehicle owners.

All detected vehicle movements, QR scans, and parking activities were stored in the MySQL database. Overall, the evaluation confirmed that ParkhAIve effectively integrated its hardware and software components to deliver accurate monitoring, reliable QR-based validation, and efficient data management.

TESTING METHODS

The researchers conducted an evaluation to assess the system's accuracy, stability, and consistency. This process is needed to identify the system's errors and weaknesses. The researchers first presented the project. Afterwards, they conducted an evaluation with the following respondents to test and assess the system's functionality.

A. Students (40)

C. Campus Security (2)



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B. ITS Employees (2)

EVALUATION CRITERIA

Below is the criteria that the respondents observed and considered during the evaluation of the system.

Table 3.4: Criteria for System Evaluation

Criteria	Indicator
Functionality	The proposed system works properly, and all expected outcomes are delivered.
	The system meets the users' work requirements, and its functions are user-friendly.
Reliability	The system can be used anytime, anywhere, without affecting its operations.
	The system operates without errors.



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Scalability	The system is adaptable and capable of meeting users' future needs.
	The system has the capability to integrate advanced hardware components.
Economic Feasibility	The system is cost-effective, and its performance is justified for its price.
	The system is built for long-term use, making it a valuable investment.
Security	The system is able to secure user data information and prevent access to non-admin accounts from the admin interface.
Maintainability	The proposed system can operate without constant maintenance.



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	The devices are carefully and precisely placed in a secure position.
Usability	Users are able to use the system without requiring a high level of digital literacy.
	The system is responsive and has minimal complications. It is compatible with various devices, ensuring accessibility for all users.

INSTRUMENT AND TECHNIQUE USED

A. Unit Test

The unit test proves that a system module executes a single unit of capabilities in accordance with a specified standard. It is an assessment of a system module's functionality alone.



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Table 3.5: Unit Test of IP Camera Detection

Test Case Name:	Camera Detection Test		TEST ID:	IPCAM-1	
Description:	Verification of the usability and functionality of IP Camera used for the parking management system				
NO:	TEST SCENARIO	EXPECTED OUTPUT	RESULT PASS/FAIL	COMMENT	
1	Frame Capture	The camera is able to CAPTURE frames from the video stream.			
2	QR Detection	The camera is able to tag the QR ID to the vehicle			
3	Vehicle Detection	Vehicle is DETECTED			
4	Data Recorded to MySQL Database	Vehicle Tagged			
		QR Pending			
		A list of all TAGGED VEHICLES is recorded			
5	Frame Capture	The camera is able to CAPTURE frames from the video stream			
6		Camera detects ‘AVAILABLE’ parking space			



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	Vacancy Detection and Designati on	Camera detects 'OCCUPIED' parking space		
		Camera processes empty parking space as 'AVAILABLE'		
		Camera processes occupied parking space as 'OCCUPIED'		
7	Vehicle Detection	Vehicle is DETECTED		
8	Slot Detection	Slots from all zones are DETECTED		
9	Data Recorded to MySQL Database	'AVAILABLE' parking space is recorded		



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Table 3.6 Unit Test for QR Code Scanning

Test Case Name:	QR Code Scanning			TEST ID:	QRCS-1
Description:	Verification of the efficiency of QR Code scanning.				
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULT PASS/FAIL	COMMENT	
1	Generate QR	QR is generated			
2	QR Scanning	QR is scanned			
3	QR Database Records	QR is recorded in the database			
4	QR ID Tagging	ID is tagged to the car after the QR is scanned			

Table 3.7: Unit Test of User Login

Test Case Name:	User Login Test			TEST ID:	ULOG-1
Description:	Verification of User Login functionality				
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULT PASS/FAIL	COMMENT	
1	If the user is not logged in	Render Basic Digital Twin			
		User creates an account and			



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		successfully inputs credentials		
2	If the user is logged in	Login successful		
		Redirects to Digital Twin Page		
3	If the user forgot their password	Redirect to Forgot Password Dialog		
4	If the user inputs incorrect credentials	Show Login Error		

Table 3.8 Unit Test of User Interface

Test Case Name:	User Interface Test			TEST ID:	UAPP-1
Description:	Verification of User Interface functionality				
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULT PASS/FAIL	COMMENT	
1	Digital Twin Rendering	Renders the Digital Twin interface of the parking lot			
2		User is able to check OCCUPIED			



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	Parking Lot Vacancy Check	parking slots		
		User is able to check the AVAILABLE parking slots		
3	Updated data recorded in the MySQL Database	Slot 'OCCUPIED' is recorded		
		Slot 'AVAILABLE' is recorded		
		Vehicle information(Ty pe, plate no., name of owner, etc.) is recorded		
4	Even Trigger Checks	Detected Vacant Slot		
		Detected name of Vehicle Owner		

Table 3.9: Unit Test of Admin Interface

Test Case Name:	Admin Interface Test			TEST ID:	AAPP-1
Description:	Verification of Admin Interface functionality				
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULT PASS/FAIL	COMMENT	



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1	Digital Twin Rendering	Renders the Digital Twin interface of the parking lot		
2	Admin Privileges Check	Admin is able to add Events on the Event Tab		
		Admin is able to generate a QR-based token for Guests		
		Admin is able to scan the user's QR Code		
		Admin is able to view all personnel who have access to the admin account		
		Admin is able to access all the Libraries for configuration		
		Admin is able to view Parking Logs		



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Table 3.10: Unit Test of Router and IP Camera Network Connection

Test Case Name:	Router and IP Camera Network Ping Connections			
Description:	Verify whether a connection is established between the Router, Switch, and IP Camera by sending a ping through the local processing server.			
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS	COMMENT
1	Router Pings	Ping from Local Server to router is successful		
2	IP Camera Pings	Ping from Local Server to IP Camera is successful		

Table 3.11 Unit Test of SMS Notifications

Test Case Name:	SMS Notifications			
Description:	Verify whether SMS is being sent and received.			
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS	COMMENT
1	OTP request	OTP is successfully sent to the user's phone number		



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2	Event Notification	Event notification is successfully sent to the user's phone number		
---	--------------------	--	--	--

B. Integration Test

Integration testing will be conducted to ascertain that all components, namely the local processing server, router, gigabit network switch, 1080p IP camera, SMS server, image processing system, machine learning algorithms, web application, and database, operate in unison as a fully integrated system in accordance with the defined requirements. This phase of testing will examine the interactions and interdependencies among these units within a real-world context, as demonstrated through a pilot implementation, thereby ensuring reliability and seamless functionality prior to large-scale deployment.

Table 3.12 Integration Test

Parameter	Description
Duration	Total time taken for the integration test session (e.g., in minutes or hours).
Camera Performance	Speed and Quality of the 1080p camera during image capture and transmission.
Image Processing Accuracy	Effectiveness of the image processing system in analyzing captured data.



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Network Conditions	Stability description of the switch, IP camera, and local processing server performance, testing their speed, stability, and latency during runtime
QR Code Scanning	Response and transmission time of QR scans to the ML system.
Database Synchronization	Speed and accuracy of data storage and retrieval between the database and other system components.
Web Application Response	Responsiveness and functionality of the web application when receiving, processing, and displaying data.
System Interoperability	Assessment of how all components communicate and function together seamlessly.
User Experience Feedback	Qualitative feedback from users on system reliability, ease of use, and performance during the test.



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parkhAlve Research Survey

Survey for parkhAlve

jees.cuenco.up@phinmaed.com [Switch account](#)



* Indicates required question

Email *



Record jees.cuenco.up@phinmaed.com as the email to be included with my response

Type of Respondent *



Student



Employees



General Services Department



Campus Security

Have you used the web application for parkhAlve? *



Yes



No



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Web Application User Experience

Functionality *

The web server is able to send SMS notifications, QR code scanning, and slot detection.

	1	2	3	4	5	
Needs improvement	☐	☐	☐	☐	☐	Outstanding

Reliability *

The system operates consistently and without errors under normal conditions.

	1	2	3	4	5	
Needs improvement	☐	☐	☐	☐	☐	Outstanding

Scalability *

The system is able to integrate advanced hardware components and expand its coverage.

	1	2	3	4	5	
Needs Improvement	☐	☐	☐	☐	☐	Outstanding

Economic Feasibility *

The system is cost-effective, requiring minimal resources for deployment and operation.

	1	2	3	4	5	
Needs improvement	☐	☐	☐	☐	☐	Outstanding



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Security *

The system is able to secure user data and prevent access to administrative privileges to users other than the admin.

	1	2	3	4	5	
Needs improvement	☐	☐	☐	☐	☐	Outstanding

Maintainability *

The system is designed for easy updates and maintenance.

	1	2	3	4	5	
Needs improvement	☐	☐	☐	☐	☐	Outstanding

Usability *

The system interface is designed to be accessible and intuitive for users with varying levels of digital literacy.

	1	2	3	4	5	
Needs improvement	☐	☐	☐	☐	☐	Outstanding

Google Forms: parkhAIve Research Survey using Likert Scale



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ANALYSIS OF DATA GATHERING METHODOLOGY

Tools for Data Analysis

The data gathered in this research were analyzed statistically using:

Weighted Mean. The weighted mean was used to analyze the data by assigning different weights to each response, ensuring a more precise evaluation of user feedback. This statistical analysis summarizes numerical inputs from user surveys and observations, placing importance on significant responses.

In this study, the weighted mean was applied to assess the system's performance based on the study's evaluation criteria and to evaluate user satisfaction from the survey results. By assigning weighted values for each response, the analysis provided a clearer and more precise representation of how respondents perceived the system's performance and effectiveness in parking slot detection and user experience. The results helped in identifying strengths and weaknesses of the system, showing potential areas for improvement and ensuring that the proposed system meets the operational needs of the users and educational institutions.

$$WM = \frac{\sum_{x=1}^n (\omega_i * x_i)}{\sum_{x=1}^n \omega_i}$$



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Where:

WM represents the weighted mean

n represents the number of ratings

ω_i = total number of responses for each rating; frequency

x_i = weighted value of each rating; Likert scale values

Percentage. This method was used to interpret how responses were distributed, helping analyze user feedback and system performance. It measured user satisfaction and overall effectiveness of the parkhAIve system. A higher percentage of positive responses would indicate higher user satisfaction and effectiveness, while a lower percentage would indicate that the following areas need improvement. This method allows the identification of user trends and understanding of the system's impact in reducing parking congestion and improving parking slot occupancy in the university campus.

To accurately convert the responses collected by the 5-point Likert Scale Survey (Outstanding, Above Satisfactory, Satisfactory, Below Satisfactory, Needs Improvement) into percentages (0%-100%), the following formula was used:

$$Percentage = \left(\frac{Weighted\ Mean}{5} \right) \times 100$$



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Where:

Percentage = percentage of responses

Weighted Mean (WM) = average score derived from Likert Scale

responses

Formula for Interval Calculation

To methodologically organize and interpret survey responses, the study utilizes the interval formula to define score classifications. This ensures a precise representation of respondent feedback.

The interval is calculated using the following formula:

$$I = \frac{\text{Highest Score} - \text{Lowest Score}}{\text{Number of Categories}}$$

Where:

I = interval

Highest Score = 5 (Outstanding)

Lowest Score = 1 (Needs Improvement)

Total Number of Categories = 5

$$I = \frac{5-1}{5} = 0.80$$

As shown in the calculations above, each category accounts for 0.80 points, providing a clear, organized method for interpreting the Likert Scale.



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Table 3.13: The Likert Scale

Rating	Scale	Range
5	Outstanding	4.21 - 5.00
4	Above Satisfactory	3.41 - 4.20
3	Satisfactory	2.61 - 3.40
2	Below Satisfactory	1.81 - 2.60
1	Needs Improvement	1.00 - 1.80



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CHAPTER IV

SYSTEM DESIGN, RESULTS, AND DISCUSSION

This chapter outlines the implementation of the parking management system, *parkhAIve*, and presents, analyzes, and interprets the findings and discussions derived from the study. It elaborates on the system's functionalities, analyzes test results, and explores user feedback in line with the study's objectives. Furthermore, it includes results presented in tables and figures, analyzed using statistical tools to evaluate the system's efficiency, usability, and overall performance.

4.1 System Design

4.1.1 Development Tools and Technologies

parkhAIve was developed using a range of hardware and software technologies to ensure efficiency, security, and usability.

- **IDE:** Visual Studio Code for code development and debugging.
- **Frontend:** React.js (with Node.js for runtime environment and installation of dependencies) for an interactive user interface.
- **Backend:** Laravel for developing the backend of the web application for authentication, server tasks, and API requests.
- **Database:** MySQL for recording and storing data of the web application.
- **Authentication:** HTTPSMS for developing an SMS server for SMS notifications in the web application.
- **Deployment:** Local Desktop Server Hosting



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- **Machine Learning Platform:** Roboflow for creating machine learning datasets.
- **Machine Learning IDE:** Cursor
- **Machine Learning Models:** OpenCV for image processing, YOLOv8n for real-time object detection, Re-Identification (ReID) model for vehicle identity matching, and ByteTrack algorithm for multi-object tracking
- **Hardware Components:**
 - **Local Processing Server Specifications:**

Laptop Specifications:

- Processor: 12th Gen Intel(R) Core(TM) i5-12450H
- Memory: 8.0 GB RAM
- GPU: NVIDIA GeForce RTX 3050 Laptop GPU
- SSD: SAMSUNG MZVL4512HBLU-00BTW

PC Specifications:

- Processor: Intel Core i7 - 8700 6 Cores, 12 Threads
- Motherboard: MSI H310 M - Pro Series
- Cooler: Cooler Master AIO ML120L V2
- Memory: 16 GB RAM 8x2 HyperX Fury
- GPU: MSI RTX 2060 Super 8GB VRAM
- SSD: 128 GB Ramsta SSD
- HDD: 1 TB HDD Seagate
- Power Supply: SuperFlower Trurated 550W



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- Case: Powerlogic Agiele White

- **1080p IP Camera w/ SD Card** - Video streaming for slot detection and ID tagging
- **TP-Link Litewave LS1008G Desktop Switch** - Network connection hub for all networking hardware components
- **ASUS Router** - Routes data between the IP camera, switch, and local processing server

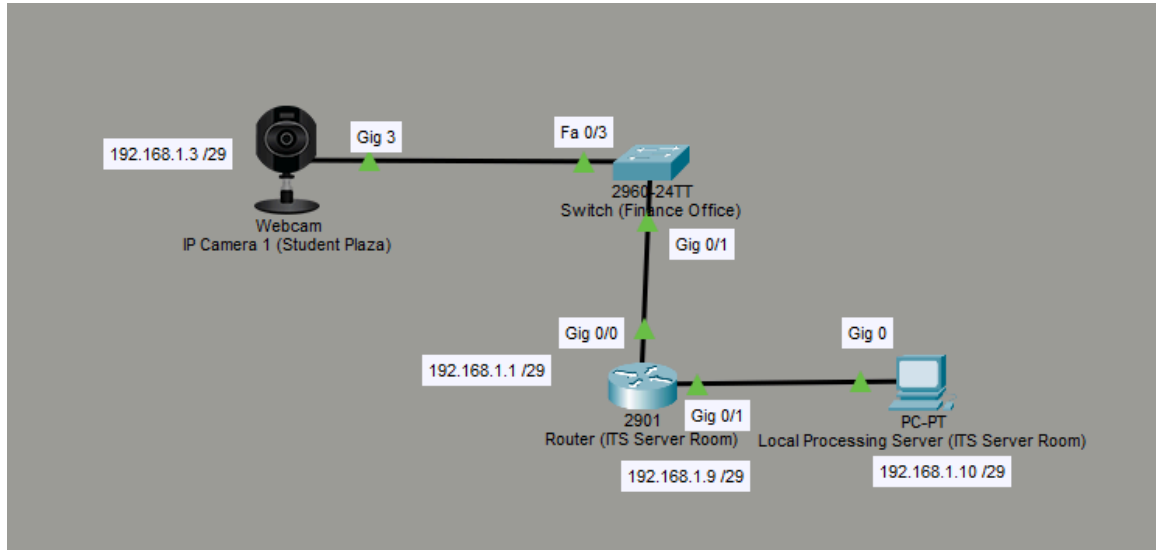
4.1.2 System Fabrication

Initial Network Configuration Setup and Testing for Hardware Components

The initial phase involved configuring the necessary hardware components, including the Desktop PC, 1080p IP Camera, Network Switch, and Router. This setup allowed for preliminary testing to validate individual component functionality, examine interactions between hardware components, and establish a blueprint for LAN cable connections between them.



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Network Configuration Diagram

Wooden Support Boards for IP Camera

Following the initial phase, wooden support boards were attached to the 1080p IP Camera to ensure stable support when deployed. This phase involves drilling holes in the wooden board and using drywall anchors and screws to secure the IP camera. Additional holes were drilled near the edges of the wooden board for steel wire support once deployment starts.



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Mounting of Wooden Support Boards for an IP Camera

Deployment of Hardware Components in the PHINMA Avenue

After the wooden support boards were made, the hardware components were deployed on PHINMA Avenue. This phase involves deploying the system's hardware components, including a 1080p IP camera, a desktop server, a router, and a network switch. The distance between hardware components was also measured for the LAN cable connections to ensure minimal excess cable lengths and reduce clutter.



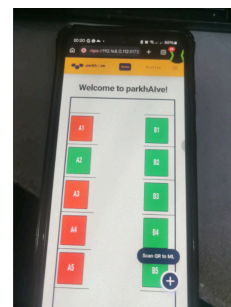
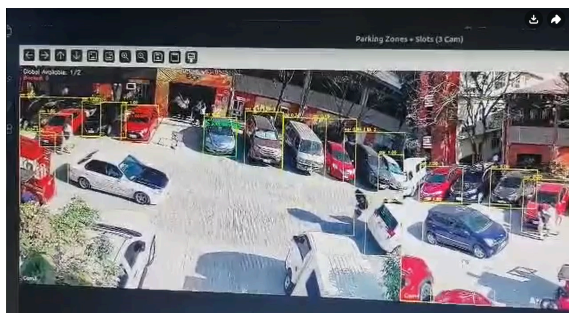
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Installing Hardware Components in the University

Integration of Hardware Components to Software

Following the deployment of hardware components, the integration between hardware and software components begins in this phase. This phase involves connecting hardware and software components and laying the foundation for the system. This phase allows testing of the system's hardware and software components, ensuring they work together seamlessly and produce minimal to no conflicts.



Integrating hardware and software components by testing the system



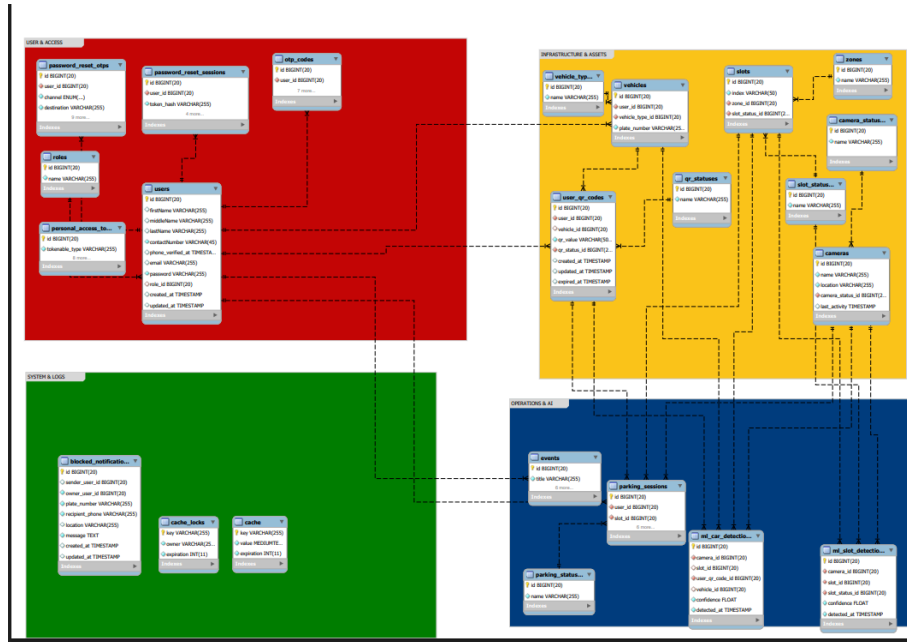
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Web Application Development

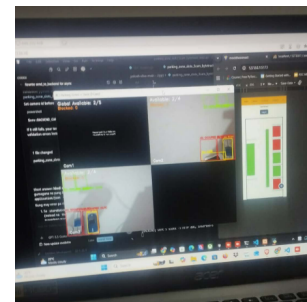
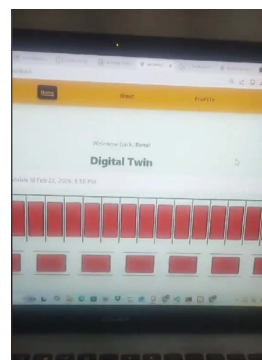
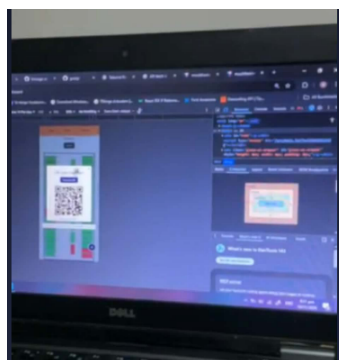
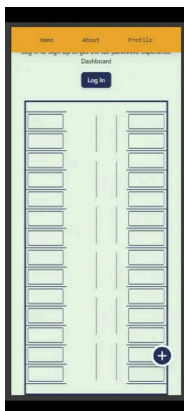
This phase involves developing the web application for the parkhAIve system. It includes the design and implementation of both the frontend (React.js) and the backend (Laravel), as well as the integration of the backend with the MySQL database. Additionally, this phase covers testing the communication between the frontend and backend through API calls and establishing the connection between the web application and the machine learning system. These processes ensure a smooth user experience and proper integration of the overall system.



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Database Design



Development of Web Application and Testing of Machine Learning for Web App

Connection



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4.1.3 System Features and Functionalities

The system is designed with the following key components:

1. Web Application for Car Owners in PHINMA Avenue.

- QR code-based ID tagging and parking slot access.
- Real-time updates on occupied and unoccupied parking slots.
- Digital twin interface for visual representation of parking area status.
- Simplistic User Interface for seamless and user-friendly experience.

2. Admin Interface

- Access to database libraries and its configuration.
- Access to Event calendars and its configuration.
- Access to web application QR scanner.
- Access to user and vehicle information for occupied parking slots.
- Authorization of admin account access control.
- Access to parking logs and arrival time records.
- Manual input overrides.
- Generation of Guest QR code for temporary access.

3. Real-time Parking Slot Monitoring using YOLOv8n Machine Learning Model

- Detection of vehicles inside the camera feed.
- Tagging of QR vehicle ID in the IP camera.
- Tracking of vehicles inside the PHINMA Avenue.



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- Slot occupancy detection through plotting slots in the parking area.

4. Evaluation and System Assessment

- User feedback data collection from 40 students, 2 tech experts, and 2 security guards.
- Technical assessments by experts on system performance.
- Usability evaluation through surveys, interviews, and observations.
- Performance monitoring based on camera performance, web application performance, user interface interactions, and technical issues.

4.1.4 System Architecture

The system follows a multi-tier architecture:

- **Presentation Layer:** A React.js frontend provides an interactive user interface for authentication, viewing parking slots, scanning QR codes, and managing profiles.
- **Logic Layer:**
 - **Web Backend Development:** Laravel processes server-side tasks of the web application, handles authentication, and performs API requests.
 - **Hardware Module:** A local processing server handles all system processes. It runs the machine learning model, establishes the connections between hardware components, sends tasks to each component, and hosts the web application server.
- **Data Layer:** MySQL stores data for parking slots, QR scans, and other user-related data, enabling real-time synchronization.



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4.1.5 System Testing and Debugging

To ensure reliability and functionality, the system underwent testing:

1. **Unit Testing:** Individual hardware components, such as 1080p IP Camera, were verified and tested to ensure expected functionality. Software components, such as database operations and machine learning, were repeatedly tested to ensure smooth operational performance.
2. **Integration Testing:** Testing the integration of hardware and software components to ensure smooth communication between frontend, backend, and hardware components, guaranteeing operational success.
3. **User Acceptance Testing (UAT):** Through the dissemination of the survey, respondents from PHINMA University of Pangasinan Dagupan Campus evaluated the system based on the study's evaluation criteria.

A **Likert-scale survey** was used to assess user experience.

- **Functionality** - A measure of the system's performance related to its expected tasks, such as QR scanning, real-time slot detection, and SMS notifications.
- **Reliability** - A measure of the system's operational stability.
- **Scalability** - Assess whether the system can be improved upon and expanded in the future.
- **Economic Feasibility** - Reviews whether the system is cost-effective and



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has long-term value.

- **Security** - Examines whether the system can protect user information and prevent unauthorized access to admin privileges.
- **Maintainability** - Determines whether the system's updates and maintenance are easily performed.
- **Usability** - Evaluates the system's user friendliness, ease of use, and ease of access.

4.1.6 Deployment Process

The system was deployed on various platforms to ensure the software is correctly configured and operational.

Software

	Web Services	Description and Usage
Frontend		React.js is used to build and develop the user interface of the parkhAlve web application.
Backend		Laravel is an open-source PHP framework that handles server-side processes, authentication, and API requests in the backend of the parkhAlve web application.



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		Node.js is an open-source, cross-platform runtime environment that enables developers to run JavaScript outside a web browser. It is used with React.js, providing a runtime environment and installing dependencies.
Database		MySQL is a database management system that uses Structured Query Language to manage and query data. It records and stores data from the parkhAIve web application.
Integrated Development Environment	Cursor	Cursor is an AI-powered code editor designed to integrate large language models into working code. It is also built as a fork in Visual Studio Code. It is used to develop the Yolov8n machine learning model for the parkhAIve system.
	VS Code	Visual Studio Code is an Integrated Development Environment (IDE) that provides a coding environment for developing software.



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Hardware

Local Processing Server		Handles all operations of the system. It is used to process parking slot occupancy, monitor vehicle entry, control system operations, and provide administrators with a live dashboard for efficient parking management.
1080p IP Camera		Captures high-resolution video and images for vehicle detection and real-time monitoring
Gigabit Network Switch		Enables seamless communication between the IP camera, the router, and the local processing server. It supports real-time video transmission, centralized data processing, and reliable network connectivity.
Router		Connects network devices and directs data between the IP camera and the local server.

Results and Discussion



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4.2.1 Data Presentation and Analysis

The study presents findings from survey responses from 44 participants, including students, employees, and security guards. The data was analyzed using Descriptive Statistics and the Likert Scale to assess the system based on the study's evaluation criteria.

Usability Testing and System Performance Evaluation

parkhAIve was initially tested by the researchers to evaluate its functionality, usability, and overall performance. These preliminary tests ensure that the system responded correctly under various conditions and that both hardware and software components operated as intended. The evaluation focused on key aspects such as slot detection, QR code processing, user login, user interface interactions, network pings, and IP camera video streaming, allowing any issues to be identified and addressed early in the process. 7 unit tests were conducted, specifically assessing (1) Camera detections, (2) QR code scanning, (3) User login, (4-5) User and admin interface, (6) Router, switch, and IP camera connection pings, and (7) SMS notifications. These tests provided a controlled environment for validating core functionality before broader testing.

Test Case Name:	IP Camera Detection Test		
Description:	Verification of the usability and functionality of the IP camera used for the parking management system		
	TEST SCENARIO	EXPECTED	RESULTS



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NO.		OUTPUT	
1	Frame Capture	The camera is able to CAPTURE frames from the video stream	PASS
2	QR Detection	The camera is able to tag the QR ID to the vehicle	PASS
3	Vehicle Detection	Vehicle is DETECTED	PASS
4	Data Recorded to MySQL Database	Vehicle Tagged	PASS
		QR Pending	PASS
		A list of all TAGGED VEHICLES is recorded	PASS
5	Frame Capture	The camera is able to CAPTURE frames from the video stream	PASS
6	Vacancy Detection and Designation	Camera detects 'AVAILABLE' parking space	PASS
		Camera detects 'OCCUPIED' parking space	PASS
		Camera processes empty parking space as 'AVAILABLE'	PASS
		Camera processes occupied parking space as 'OCCUPIED'	PASS
7	Vehicle Detection	Vehicle is DETECTED	PASS



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8	Slot Detection	Slots from all zones are DETECTED	PASS
9	Data Recorded to MySQL Database	‘AVAILABLE’ parking space is recorded	PASS
		‘OCCUPIED’ parking space is recorded	PASS
		A list of all DETECTED VEHICLES is recorded	PASS

Table 4.1 Unit Test of IP Camera Detection

Findings of Table 4.1 - IP Camera Detection Test

Table 4.1. The camera detection tests verified the overall usability and functionality of the IP camera within the parking management system. Results show that the camera successfully captured frames from the video stream, accurately detected QR codes, and effectively associated them with corresponding vehicles. Additionally, the system reliably identified vehicles and correctly determined parking slot occupancy by designating spaces as “AVAILABLE” or “OCCUPIED.” All relevant data were properly recorded in the MySQL database, including tagged vehicles, pending QR entries, parking space statuses, and the complete list of detected vehicles. The system consistently demonstrated accurate vehicle detection, QR-to-vehicle association, and slot identification across all



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defined zones. Overall, all test scenarios passed successfully, confirming the effectiveness of the camera in supporting both identification and monitoring processes within the system.

Test Case Name:	QR Code Scanning		
Description:	Verification of the efficiency of QR Code scanning.		
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS
1	Generate QR	QR is generated	PASS
2	QR Scanning	QR is scanned	PASS
3	QR Database Records	QR is recorded in the database	PASS
4	QR ID Tagging	ID is tagged to the car after the QR is scanned	PASS

Table 4.2 Unit Test of QR Code Scanning

Findings of Table 4.2 Unit Test of QR Code Scanning

Table 4.2. The QR Code Scanning functionality was successfully verified. The system was able to generate QR codes, scan them efficiently, record the QR



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data into the database, and correctly tag the associated ID to the vehicle after scanning, with all test scenarios passing as expected.

Test Case Name:	User Login		
Description:	Verification of User Login functionality		
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS
1	If the user is not logged in	Render Basic Digital Twin	PASS
		User creates an account and successfully inputs credentials	PASS
		Login successful	PASS
2	If the user is logged in	Redirects to Digital Twin Page	PASS
3	If the user forgot their password	Redirect to Forgot Password Dialog	PASS
4	If the user inputs incorrect credentials	Show Login Error/Incorrect Password	PASS

Table 4.3 Unit Test of User Login



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Findings of Table 4.3 - User Login

Table 4.3. The User Login functionality was successfully verified. When a user is not logged in, the system renders the basic Digital Twin interface. Users can create an account, input credentials, and log in successfully. Logged-in users are redirected to the Digital Twin page. The system also correctly handles forgotten passwords by redirecting to the Forgot Password dialog and displays an error message when incorrect credentials are entered, with all scenarios passing as expected.

Test Case Name:	User Interface		
Description:	Verification of User Interface functionality		
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS
1	Digital Twin Rendering	Renders the Digital Twin interface of the parking lot	PASS
2	Parking Lot Vacancy Check	User is able to check OCCUPIED parking slots	PASS
		User is able to check AVAILABLE parking slots	PASS
3	Updated data recorded in the MySQL Database	Slot 'OCCUPIED' is recorded	PASS



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		Slot 'AVAILABLE' is recorded	PASS
		Vehicle information (Type, plate no., name of owner, etc.) is recorded	PASS
4	Event Trigger Checks	Detected vehicle parked in front of user	PASS
		Detected Vacant Slot	PASS
		Detected name of Vehicle Owner	PASS

Table 4.4 Unit Test of User Interface

Findings of Table 4.4 - User Interface

Table 4.4. The User Interface functionality was successfully verified. The system accurately rendered the Digital Twin interface of the parking lot, allowed users to check both occupied and available parking slots, and correctly recorded updated data in the MySQL database, including slot status and vehicle information such as type, plate number, and owner's name. Event triggers also operated as expected, detecting parked vehicles, vacant slots, and vehicle owner names, with all tests passing.

Test Case Name:	Admin Interface
Description:	Verification of Admin Interface functionality



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NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS
1	Digital Twin Rendering	Renders the Digital Twin interface of the parking lot	Pass
2	Admin Privileges Check	Admin is able to add Events on the Event Tab	Pass
		Admin is able to generate a QR-based token for Guests	Pass
		Admin is able to scan the user's QR Code	Pass
		Admin is able to view all personnel who have access to the admin account	Pass
		Admin is able to access all the Libraries for configuration	Pass
		Admin is able to view Parking Logs	Pass
3	Event Trigger Checks	Detected Vehicle parked in front of user	Pass
		Detected Vacant Slot	Pass
		Detected Name of vehicle owner	Pass

Table 4.5 Unit Test of Admin Interface



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Findings of Table 4.5- Admin Interface

Table 4.5. The system successfully rendered the Digital Twin interface of the parking lot. Admin functionalities were fully verified, with the administrator able to add events, generate QR-based tokens for guests, scan user QR codes, view personnel with admin access, access configuration libraries, and review parking logs, all passing as expected. Additionally, event triggers accurately detected vehicles parked, identified vacant slots, and recognized vehicle owners, confirming proper system operation.

Test Case Name:	Router and IP Camera Network Connection		
Description:	Verify whether a connection is established between the Router and the IP Camera by sending a ping test through the local processing server.		
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS
1	Router Pings	Ping from Local Server to router is successful	Pass
2	IP Camera Pings	Ping from Local Server to IP Camera is successful	Pass

Table 4.6 Unit Test of Router and IP Camera Network Connection

Findings of Table 4.6 - Router and IP Camera Network Connection



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Table 4.6. The network connectivity test for the Router and IP Camera verified that all devices could successfully communicate with the local processing server via ping. The server was able to ping the Router and IP camera with all tests passing as expected.

Test Case Name:	SMS Notifications		
Description:	Verify whether SMS is being sent and received.		
NO.	TEST SCENARIO	EXPECTED OUTPUT	RESULTS
1	OTP request	OTP is successfully sent to the user's phone number	Pass
2	Event Notification	Event Notification is successfully sent to the user's phone number.	

Table 4.7 Unit Test of SMS Notifications

Findings of Table 4.7 - SMS Notifications

Table 4.7. The SMS Notifications test case verifies whether SMS messages are correctly sent and received. During testing, OTP requests and Event Notification were successfully delivered to users' numbers, with all scenarios passing as expected.



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Integration Testing

The parkhAIve underwent a pilot test after deployment was completed during the fabrication phase to simulate real-world conditions. Unlike the unit testing conducted earlier, integration testing is designed to verify that all hardware and software components are integrated and functioning in unison seamlessly during test sessions. Critical component functionalities such as QR code scanning, camera performance during detection, user authentication, admin interface accessibility, and web application responsiveness were accessed in a live environment. By simulating real-world scenarios, the pilot integration tests provided the researchers with valuable insights into the system's actual performance, uncovering potential issues and bugs, and ensuring the system is reliable and functional.

Parameters	Pilot Test	Unit Test
Duration	10 minutes (QR Scanning not involved due to unforeseen circumstances during the pilot testing)	30-45 minutes 3
Camera Performance	Minimal Delay (Due to ML Processing and Hardware Limitation), 720p 10fps video quality	Slight Delay (due to ML processing), 1080p video quality
Image Processing Accuracy	75 percent	75 percent



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Network Conditions (1080p Cameras to Local Processing Server)	Stable (Cat6 & 1GBps)	Stable
QR Code Scanning	2-5 seconds (after scan and transmitted to ML)	2-5 seconds (after scan and transmitted to ML)
Database Synchronization	Instantaneous	Instantaneous
Web Application Response	10 - 15 seconds, 13.5 second average	10 - 15 seconds, 13.5 second average
System Interoperability	All system components communicated and worked together seamlessly, enabling smooth data exchange between the camera, the database, and the web application.	N/A
User Experience Feedback	The web app is easy to navigate and understand. UI is not clustered. Digital twins are easy to follow and understand.	N/A

During the pilot and unit testing phases, the system demonstrated consistent performance across several of the key parameters in the table. The pilot test was completed for about 10 minutes, while the unit test was completed in 45 minutes. This gap in duration shows a much faster test time due to various fixes and optimizations during the development of the system. In terms of camera performance, the pilot tests show a slight delay at 720p 30fps video quality



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compared to the unit tests' slight delay on 1080p quality. The reduction in quality is due to hardware limitations and machine learning processes.

The rest of the key parameters show similar results across all pilot and unit tests. The image processing accuracy shows 75% accuracy in detecting and processing data between the camera and the local processing server. Database synchronization shows instantaneous responses. Web application responses show an average of about 13.5 seconds. This rather long response time is due to multiple considerations related to the machine learning system and its slot detection functionality. Network conditions are stable in all tests, and the QR code scanning takes an average of around 2 - 5 seconds on all tests to transmit and process to the local processing server. The system interoperability tests have confirmed that all system components are communicating and working together seamlessly.

Interview Results and Summary

An interview with a software expert, Jaby Bautista, provided the researchers with valuable and constructive insights and recommendations regarding the parkhAIve system.

Sir Jaby Bautista, an Information and Technology Services Department Employee at PHINMA UPang, noted that the parkhAIve system can improve



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campus parking efficiency and security. However, concerns were raised regarding long-term sustainability, particularly in handling technical support and maintenance after deployment. The need for clear support from the institution and consideration of both initial and ongoing costs was emphasized. The expert also recommended adding an admin notification feature for unscanned vehicles parking within PHINMA Avenue and continuously upgrading the server specifications to ensure smoother and faster video processing over time.

Additionally, an interview with a campus security guard, Guard Dacanay, provided the researchers with additional insights about the system's capabilities and limitations, emphasizing digital literacy for older people and proper dissemination.

Guard Dacanay of PHINMA UPang stated that the system is well-developed and would greatly reduce their workload, particularly in manual checking of parking availability and recording visitor passes for parking along the designated monitoring zone on PHINMA Avenue. However, he emphasized that if the institution adopts the system, proper information dissemination is necessary to prevent traffic congestion at the parking entrance. He also noted that employees and visitors aged 45 and above may require sufficient digital literacy to effectively use the system.



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The combined insights of the IT professional and Guard Dacanay show that the proposed smart parking system is an effective solution for improving efficiency and reducing manual workload. However, its success depends on proper technical support, cost planning, system upgrades, and clear information dissemination. Additionally, addressing users' digital literacy is essential to ensure smooth and inclusive implementation.

User Evaluation Results (Likert Scale 1-5):

Functionality The web server is able to send SMS notifications, QR code scanning, and slot detection.			
Likert Scale	Likert Scale values(x_i)	Frequency(ω_i)	Mean ($\omega_i * x_i$)
Outstanding	5	21	105
Above Satisfactory	4	11	44
Satisfactory	3	8	24
Below Satisfactory	2	2	4
Needs Improvement	1	2	2
		Weighted Mean	4.068

The functionality of the system received a weighted mean of 4.068, which is rated as Above Satisfactory. This means users found the system to be effective in performing the tasks it is expected to perform, such as parking slot detection, QR code scanning, and SMS notifications.



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Reliability The system operates consistently and without errors under normal conditions.			
Likert Scale	Likert Scale values (x_i)	Frequency(ω_i)	Mean ($\omega_i * x_i$)
Outstanding	5	12	60
Above Satisfactory	4	19	76
Satisfactory	3	9	27
Below Satisfactory	2	2	4
Needs Improvement	1	2	2
		Weighted Mean	3.840

The reliability of the system received a weighted mean of 3.840, which is rated as Above Satisfactory. This means users found the system to be reliable in detecting parking slots, QR code scans, and SMS notifications.

Scalability The system is able to integrate advanced hardware components and expand its coverage.			
Likert Scale	Likert Scale values (x_i)	Frequency(ω_i)	Mean ($\omega_i * x_i$)
Outstanding	5	13	65
Above Satisfactory	4	24	96
Satisfactory	3	3	9



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Below Satisfactory	2	2	4
Needs Improvement	1	2	2
		Weighted Mean	4.00

The scalability of the system received a weighted mean of 4.00, which is rated as Above Satisfactory. This means users find the system to be scalable in hardware, and capable of developing with more advanced hardware for better performance and area coverage.

Economic Feasibility The system is cost-effective, requiring minimal resources for deployment and operation.			
Likert Scale	Likert Scale values (x_i)	Frequency(ω_i)	Mean ($\omega_i * x_i$)
Outstanding	5	15	75
Above Satisfactory	4	21	84
Satisfactory	3	3	9
Below Satisfactory	2	3	6
Needs Improvement	1	2	2
		Weighted Mean	4.00

The economic feasibility of the system received a weighted mean of 4.00, which is rated as Above Satisfactory. This means that users find the system to be cost-effective, its resource usage being justified in terms of its performance.



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Security The system is able to secure user data and prevent access to administrative privileges to users other than the admin.			
Likert Scale	Likert Scale values (x_i)	Frequency(ω_i)	Mean ($\omega_i * x_i$)
Outstanding	5	14	70
Above Satisfactory	4	20	80
Satisfactory	3	6	18
Below Satisfactory	2	2	4
Needs Improvement	1	2	2
		Weighted Mean	3.954

The security of the system received a weighted mean of 3.954, which is rated as Above Satisfactory. This means users find the system to be secure in handling and protecting user information, and prevent non-admin accounts from accessing the web application's admin interface.

Maintainability The system is designed for easy updates and maintenance.			
Likert Scale	Likert Scale values (x_i)	Frequency(ω_i)	Mean ($\omega_i * x_i$)
Outstanding	5	15	75
Above Satisfactory	4	18	72
Satisfactory	3	7	21
Below Satisfactory	2	2	4



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Needs Improvement	1	2	2
		Weighted Mean	3.954

The maintainability of the system received a weighted mean of 3.954, which is rated as Above Satisfactory. This means users find the system to be very easy to maintain and perform updates on features and optimizations.

Usability The system interface is designed to be accessible and intuitive for users with varying levels of digital literacy.			
Likert Scale	Likert Scale values (x_i)	Frequency(ω_i)	Mean ($\omega_i * x_i$)
Outstanding	5	14	70
Above Satisfactory	4	22	88
Satisfactory	3	4	12
Below Satisfactory	2	2	4
Needs Improvement	1	2	2
		Weighted Mean	4.00

The usability of the system received a weighted mean of 4.00, which is rated as Above Satisfactory. This means users find the system's web application to be accessible and user-friendly. The weighted mean results show that the web interface is incredibly intuitive and easy to use for varying levels of digital literacy.



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Factors	Weighted Mean	Percentage
Functionality	4.068	81.36%
Reliability	3.840	76.80%
Scalability	4.00	80.00%
Economic Feasibility	4.00	80.00%
Security	3.954	79.08%
Maintainability	3.954	79.08%
Usability	4.00	80.00%

Table 4.8 Percentage Table

USER EVALUATION FINDINGS AND SUMMARY

From the results shown above, the parkhAIve system achieved a total combined weighted mean of 4.238, which puts the system at an overall rating of Outstanding on the Likert Scale. This result shows that the users were satisfied with the performance of the system in terms of parking slot detection, web application experience, and overall usability. The use of the Yolov8n machine learning model is an effective tool to improve parking slot occupancy and vehicle detection. QR code scanning also gives convenient, secure and organized access to parking slots to keep the parking area of the university in order as well as improve user interaction. The web application is also an effective way to provide



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easy-to-understand information about the parking area of the university without having to be physically present.

The Likert Scale provides researchers a clear and structured way to measure user feedback, ensuring that the results are based on real data. This data not only proves that the system is effective but also defines the areas that are in need of improvement in the future. The percentage table shows the overall satisfaction of the respondents in using the parkhAIve system in percentage form. The table has shown a decently high percentage rating across all factors, this suggests that the users have a great number of positive responses when using the system.



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CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary of Findings

This study aimed to develop a YOLOv8n-based Smart Parking Management System that monitors parking slot availability in real-time through the YOLOv8n machine learning model and 1080 IP Camera. The system integrates a Progressive Web Application (PWA) for easy access to parking slot information, organized parking management through QR code scanning, user authentication for secure user data and access, and an admin interface for easy information updates and access to parking logs and event libraries.

The key findings of the study are as follows:

- The system successfully determines parking slot occupancies in the web application through 1080p IP Camera and the YOLOv8n machine learning



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vehicle detection model with the aid of ByteTrack, OpenCV, and ReID, improving parking efficiency and parking slot monitoring.

- The system effectively organizes parking slot access through QR Code scanning, reducing parking congestion and adding a layer of security within the parking area of PHINMA University of Pangasinan.
- The web application-based digital twin interface is an effective and easy way for users to inquire information about parking slot status in the university campus. It also allows administrators (such as security guards, employees, or other admin-qualified personnel) an efficient and quick way to view the status of the parking area in one place.
- User feedback gathered from students, employees, and security guards indicated that the system is efficient, effective, user-friendly, and reliable in performing its tasks.
- Testing methodologies, including unit testing, integration testing, and user acceptance testing (UAT), confirmed that the system is functional, reliable, and effective.
- The system evaluation was based on key criteria such as functionality, reliability, scalability, economic feasibility, security, maintainability, and usability.



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5.2 Conclusions

The findings demonstrate that the parkhAIve system provides a functional solution for parking slot availability monitoring. By utilizing the YOLOv8n model and a 1080p IP camera, the system was able to perform real-time vehicle detection and parking slot classification. The developed web-based interface allows users to view parking slot information in real time, improving accessibility and user interaction within the system.

After thorough testing and evaluation, the system has been proven to be technically feasible, user-friendly, and adaptable, with the proper support of the institution to meet the needs of educational establishments. The feedback collected from various respondents further supports the system's potential as a practical and scalable solution for parking management, especially within the university campus of PHINMA University of Pangasinan.

5.3 Recommendations

To enhance the ParkhAIve system, it is recommended that future researchers focus on improving system accuracy, usability, and reliability. This includes further training of the machine learning model using more diverse datasets to enhance detection performance under varying environmental conditions.

Additionally, the development of a user-friendly interface or mobile-based access for real-time parking updates may improve user convenience without requiring full



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automation. Refining the hardware configuration and the web interface will create a more intuitive web application for both administrators and users, especially as the system should be designed to cover the entire PHINMA Avenue parking space.

The system may also benefit from incorporating backup and fail-safe mechanisms to ensure continuous operation during technical issues. Upgrading the local processing server with higher RAM, storage, and GPU specifications is also recommended to improve overall system performance and support more efficient real-time processing.

Lastly, future researchers may explore expanding the system's coverage and seeking institutional support to optimize its performance for larger parking areas upon deployment, while maintaining its cost-effective and non-intrusive design.

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PHINMA UNIVERSITY OF PANGASINAN

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PHINMA UNIVERSITY OF PANGASINAN

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APPENDICES



PHINMA UNIVERSITY OF PANGASINAN

APPENDIX A

SURVEY

parkhAlve Research Survey

Survey for parkhAlve

* Indicates required question

Email *

☐ Record team.gonatic.up@phinmaed.com as the email to be included with my response

Type of Respondent *

☐ Student

☐ Employees

☐ Campus Security

Have you used the web application for parkhAlve? *

☐ Yes

☐ No

[Next](#) [Clear form](#)

parkhAlve Research Survey

Your email will be recorded when you submit this form

* Indicates required question

Web Server User Experience

Functionality *

The web server is able to send SMS notifications, QR code scanning, and slot detection.

1 2 3 4 5

Needs improvement ☐ ☐ ☐ ☐ ☐ Outstanding

Reliability *

The system operates consistently and without errors under normal conditions.

1 2 3 4 5

Needs improvement ☐ ☐ ☐ ☐ ☐ Outstanding

Extensibility *

The system is able to integrate advanced hardware components and expand its coverage.



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APPENDIX A

SURVEY

Extensibility *

The system is able to integrate advanced hardware components and expand its coverage.

12345

Needs improvement

☐

☐

☐

☐

☐

Outstanding

Economic Feasibility *

The system is cost-effective, requiring minimal resources for deployment and operation.

12345

Needs improvement

☐

☐

☐

☐

☐

Outstanding

Security *

The system is able to secure user data and prevent access to administrative privileges to users other than the admin.

12345

Needs improvement

☐

☐

☐

☐

☐

Outstanding

Maintainability *

The system is designed for easy updates and maintenance.

12345

Needs improvement

☐

☐

☐

☐

☐

Outstanding

Maintainability *

The system is designed for easy updates and maintenance.

12345

Needs improvement

☐

☐

☐

☐

☐

Outstanding

Usability *

The system interface is designed to be accessible and intuitive for users with varying levels of digital literacy.

12345

Needs improvement

☐

☐

☐

☐

☐

Outstanding

☐ Send me a copy of my responses.

Back

Submit

Clear form

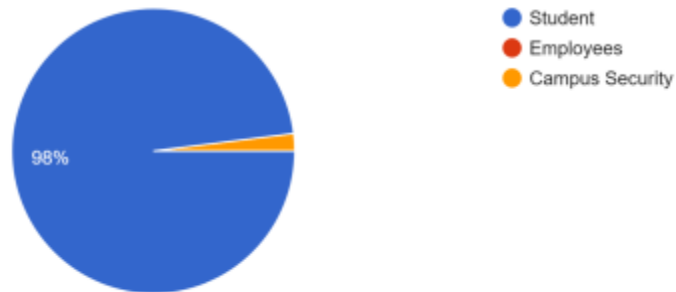
126



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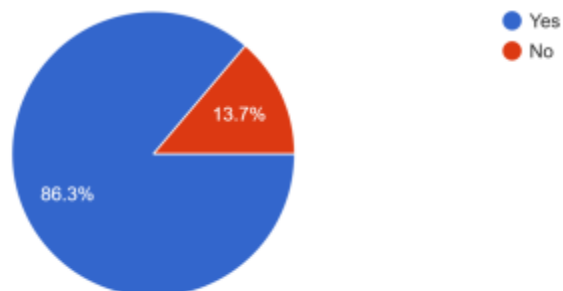
Type of Respondent

44 responses



Have you used the web application for parkhAlve?

44 responses

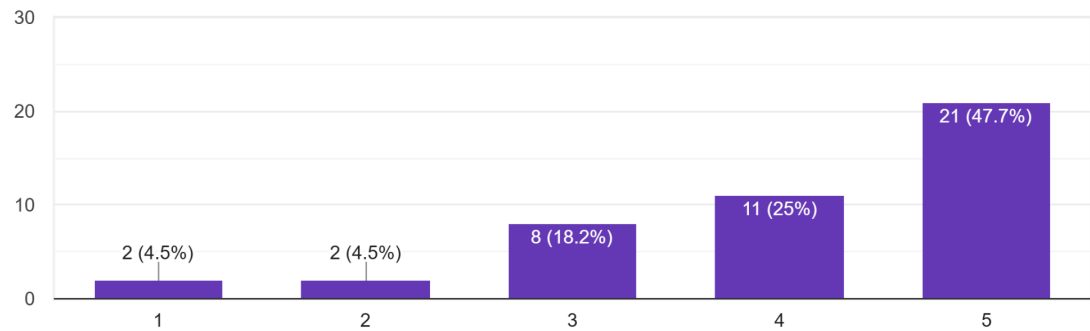




PHINMA UNIVERSITY OF PANGASINAN

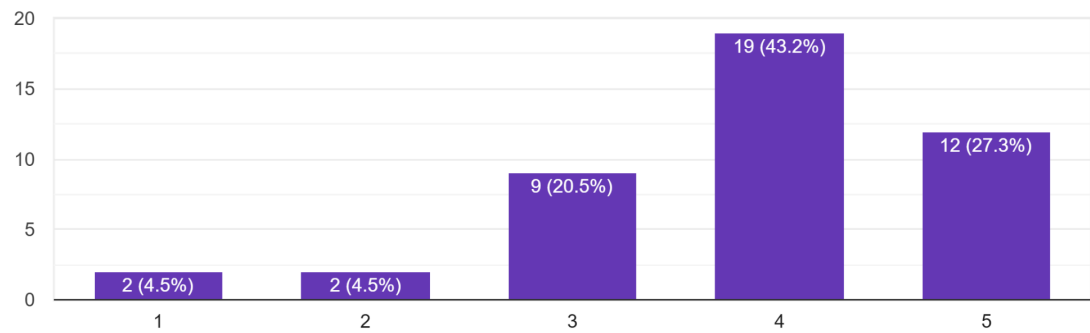
Functionality

44 responses



Reliability

44 responses

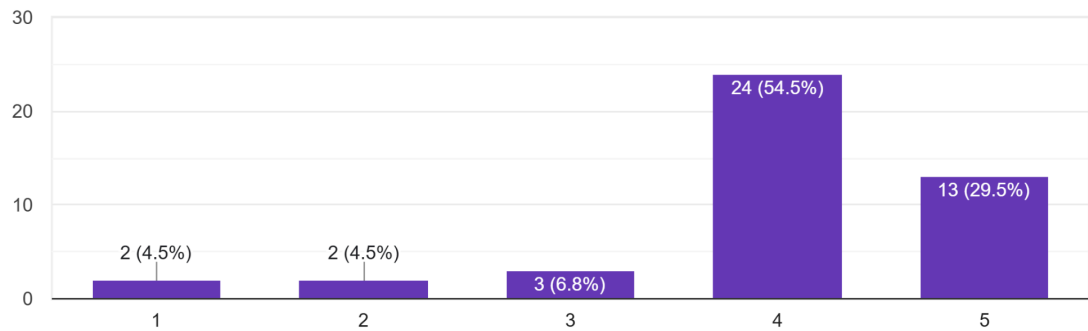




PHINMA UNIVERSITY OF PANGASINAN

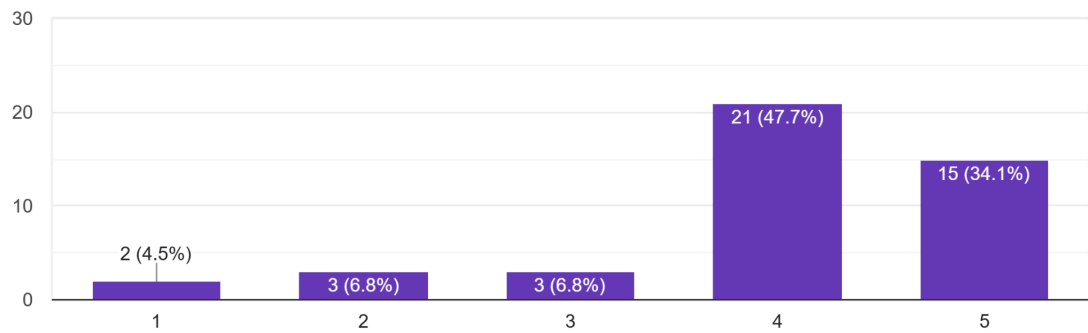
Extensibility

44 responses



Economic Feasibility

44 responses

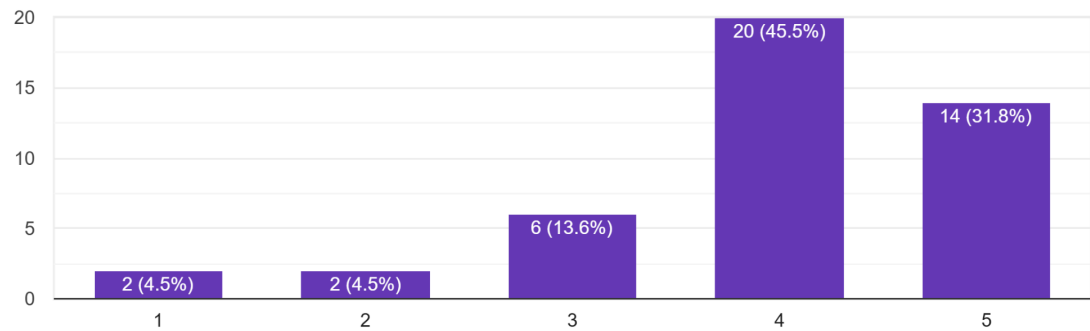




PHINMA UNIVERSITY OF PANGASINAN

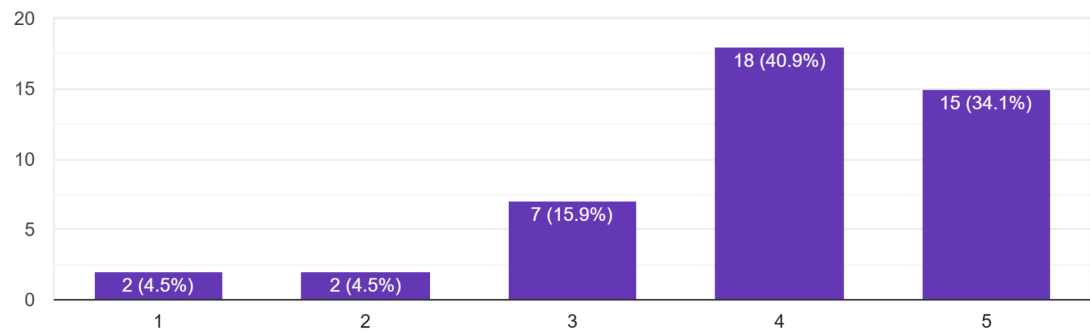
Security

44 responses



Maintainability

44 responses

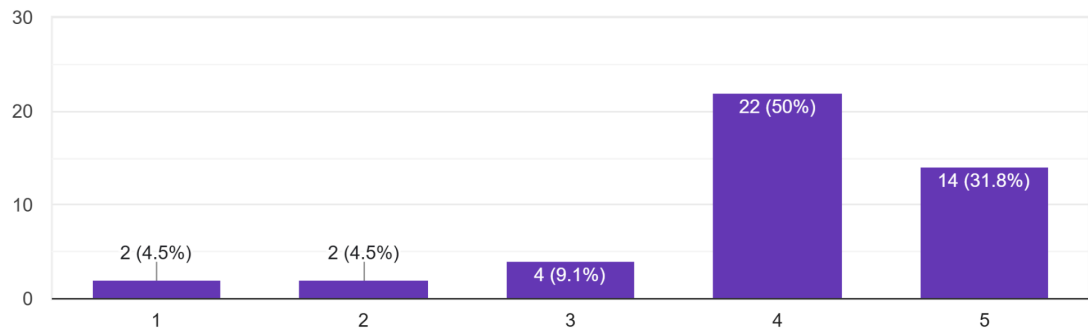




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Usability

44 responses





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APPENDIX B

CURRICULUM VITAE

RANA LOUISE S. CARLOS

Villa Sta. Barbara, Sta. Barbara, Pangasinan

rasi.carlos.up@phinmaed.com



EDUCATION

PHINMA UNIVERSITY OF PANGASINAN

College – Bachelor of Science in Computer Engineering

2022 – Present

WORK EXPERIENCE

ITS Intern

Region 1 Medical Center MIS Department

Dagupan City, Pangasinan

May 2025 – June 2025

- Provided IT support for troubleshooting hardware and software issues.
- Developed a web-based application for the Animal Bite Treatment Center
- Performed general office duties such as documentation and data encoding.

SKILLS AND INTERESTS

Technical Skills

Programming & Development: C++, Python, Java, JavaScript, PHP, HTML, CSS, MySQL

Networking: Cisco Packet Tracer

Tools: Github, Visual Studio Code, Canva, Adobe Photoshop, Figma

Soft Skills

- Strong communication and collaboration skills
- Effective time management and multitasking
- Adaptable and eager to learn
- Attention to detail
- Problem solving and critical thinking abilities

Languages: English (Fluent), Filipino (Fluent)

Interests: UI/UX, Software Development, Web Development

LEADERSHIP

Auditor

Institute of Computer Engineers of the Philippines Student Edition – PHINMA University of Pangasinan Chapter

June 2024 – April 2025



PHINMA UNIVERSITY OF PANGASINAN

SEMINARS, TRAININGS, AND ACTIVITIES

Smart City and the Future of Jobs: Singapore Perspective

By Dr. Ryan S. Evangelista – 2025

Application Security: OWASP Top 10 – 2021

By Michael Ian Rabara – 2025

AI and Cybersecurity

By Tzar C. Umang – 2025

Data Privacy For All: Protecting Your Data, Safeguarding Your Future

By Roberth S. Lloren – 2025

AWARDS & RECOGNITION

Pin Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2025)

Hard Hat Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2024)



PHINMA UNIVERSITY OF PANGASINAN

APPENDIX B

CURRICULUM VITAE

JED DANIEL E. CUENCO

Lucao District, Dagupan City, Pangasinan
jees.cuenco.up@phinmaed.com



EDUCATION

PHINMA UNIVERSITY OF PANGASINAN

College – Bachelor of Science in Computer Engineering

2022 – Present

CERTIFICATION

Civil Service Eligibility — Professional Level

Civil Service Commission, October 2025

Electrical Installation and Maintenance National Certificate II (EIM NC II)

Technical Education and Skills Development Authority – 2024

WORK EXPERIENCE

On-The-Job Trainee

Department of Information and Communications Technology Region 1

San Fernando City, La Union, Philippines

May 2025 – June 2025

- Managed monitoring and documentation for public Wi-Fi sites to ensure consistent service delivery and reliability.
- Facilitated daily administrative operations by organizing project files.
- Executed network diagnostics and authored technical reports to streamline ICT infrastructure enhancements.

SKILLS AND INTERESTS

Technical Skills

Programming & Development: Python, C++, Java, HTML, CSS, GDScript

Electronics: Circuit and PCB Design, Troubleshooting, Soldering and Assembly of Components

Tools: Cisco Packet Tracer, Proteus, AutoCAD

Soft Skills

- Time Management
- Communication
- Teamwork
- Creativity



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- Attention to Detail
- Multitasking
- Adaptability

Languages: English (Fluent), Filipino (Fluent)

Interests: Networking, Open-source software and hardware systems, Mechatronics, Software Development

SEMINARS, TRAININGS, AND ACTIVITIES

Internet of Things Conference 2025

Packetworx – September 10, 2025

Smart City and the Future of Jobs: Singapore Perspective

By Dr. Ryan S. Evangelista – 2025

Application Security: OWASP Top 10 – 2021

By Michael Ian Rabara – 2025

AI and Cybersecurity

By Tzar C. Umang – 2025

Data Privacy For All: Protecting Your Data, Safeguarding Your Future

By Roberth S. Lloren – 2025

Getting Started with Arduino: From Components to Code

By Jasmin Kate Patinyo - 2026

AWARDS & RECOGNITION

Pin Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2025)

Hard Hat Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2024)



PHINMA UNIVERSITY OF PANGASINAN

APPENDIX B

CURRICULUM VITAE

TERRYLLE AUGIE A. GONATICE

Puelay, Villasis, Pangasinan

team.gonatice.up@phinmaed.com

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EDUCATION

PHINMA UNIVERSITY OF PANGASINAN

College – Bachelor of Science in Computer Engineering

2022 – Present

CERTIFICATION

Civil Service Eligibility — Professional Level

Civil Service Commission, October 2025

WORK EXPERIENCE

On-The-Job Trainee

Department of Information and Communications Technology Region 1

San Fernando City, La Union, Philippines

May 2025 – June 2025

- Monitored and documented Free Wi-Fi for All program sites to ensure operational reliability and public accessibility.
- Assisted in promoting digital literacy and data privacy awareness through livestreaming support and webinar content creation.
- Participated in network diagnostics and technical documentation to support ICT infrastructure improvements.

SKILLS AND INTERESTS

Technical Skills

Networking & Cybersecurity: Cisco Packet Tracer, Nmap, Wireshark, Networking Fundamentals, Cybersecurity Fundamentals, OWASP Top 10

Operating Systems & Tools: Windows, Linux, Git, GitHub, Command Line Interface, Basic IT Systems, Software Troubleshooting

Soft Skills

- Adaptability in dynamic environments
- Attention to detail
- Effective teamwork & collaboration
- Clear and concise communication
- Proactive leadership



PHINMA UNIVERSITY OF PANGASINAN

- Strategic problem-solving
- Curiosity-driven learner with a strong drive for continuous improvement

Languages: English (Fluent), Filipino (Fluent)

Interests: Cybersecurity, Digital Forensics, Networking, CTF, Leadership

LEADERSHIP

Student Development Officer September 2025 – Present
Institute of Computer Engineers of the Philippines – Student Edition Region 1 Chapter

Vice Governor June 2025 – Present
College of Engineering and Architecture Student Council, PHINMA University of Pangasinan

Public Information Officer June 2024 – April 2025
College of Engineering and Architecture Student Council, PHINMA University of Pangasinan

Institute of Computer Engineers of the Philippines (ICpEP) Representative June 2023 – April 2024
College of Engineering and Architecture Student Council, PHINMA University of Pangasinan

SEMINARS, TRAININGS, AND ACTIVITIES

Internet of Things Conference 2025
Packetworx – September 10, 2025

Blue Team Junior Analyst
Security Blue Team – November 2, 2025

Networking Basics
Cisco Networking Academy – June 5, 2025

Google Cybersecurity Certificate
Google via Coursera – April 3, 2025

Vulnerability Management – Foundation
Qualys – February 28, 2025

Cyber Apocalypse CTF 2025: Tales from Eldoria
Hack The Box – March 21-26, 2025

AWARDS & RECOGNITION

Pin Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2025)

Leadership Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2023-2025)

Leadership Awardee – The Golden Flame, PHINMA University of Pangasinan (2025)

PEN Values Awardee: Patriotism – The Golden Flame, PHINMA University of Pangasinan (2025)

Hard Hat Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2024)

Leadership Awardee – PHINMA UPang Student Leaders' Synergy and Excellence (PULSE) Awards (2024)



PHINMA UNIVERSITY OF PANGASINAN

APPENDIX B

CURRICULUM VITAE

SANTIAGO C. NAVATO II

Poblacion, Bugallon, Pangasinan

saco.navato.up@phinmaed.com

www.linkedin.com/in/santiago-navato-527a57353



EDUCATION

PHINMA UNIVERSITY OF PANGASINAN

College – Bachelor of Science in Computer Engineering

2022 – Present

WORK EXPERIENCE

On-The-Job Trainee

Department of Information and Communications Technology Region 1

San Fernando City, La Union, Philippines

May 2025 – June 2025

- Streamlined administrative workflows for the ILCDB and other projects/sub-departments, significantly increasing operational efficiency through meticulous documentation and process management.
- Assisted in the Free Wifi for All department in monitoring and updating status of internet connections throughout Region 1.
- Conducted on site visitations to different government areas in checking internet connections with the GovNet team.
- Facilitated seminars in collaboration with the ILCDB, providing on-site support and managing interactive sessions.

SKILLS AND INTERESTS

Technical Skills

Programming & Development: Proficient in C++, Java, and Python (basic level)

Electronics: Experienced in circuit soldering, circuit design, and Arduino-based hardware integration.

Tools: Cisco Packet Tracer, Capcut, Canva

Soft Skills

- Strong time management
- Adaptability
- Effective leadership capabilities
- Ability to work independently or collaboratively
- Decision-making under pressure

Languages: English (Fluent), Filipino (Fluent), Pangasinan (Fluent)

Interests: Leadership



PHINMA UNIVERSITY OF PANGASINAN

LEADERSHIP

College of Engineering and Architecture Representative June 2025 – Present
Supreme Student Council, PHINMA University of Pangasinan

Vice President for External Affairs August 2025 – Present
Institute of Computer Engineers of the Philippines – Student Edition Region 1 Chapter

Vice President for External Affairs June 2024 – April 2025
Institute of Computer Engineers of the Philippines Student Edition – PHINMA UPang Chapter

Editor August 2022 – April 2024
Supreme Student Council, PHINMA University of Pangasinan

SEMINARS, TRAININGS, AND ACTIVITIES

Internet of Things Conference 2025
Packetworx – September 10, 2025

Hack The Chain Ep25.2
Makerspace InnovHub – July 9-13, 2025

ICpEP 7th Regional Convention and CPE Challenge
Institute of Computer Engineers of the Philippines Region 1 Chapter – April 28, 2025

Smart City and the Future of Jobs: Singapore Perspective
Dr. Ryan S. Evangelista – 2025

Application Security: OWASP Top 10 – 2021
Michael Ian Rabara – 2025

AI and Cybersecurity
Tzar C. Umang – 2025

Data Privacy For All: Protecting Your Data, Safeguarding Your Future
Roberth S. Lloren – 2025

Data Privacy Awareness
DICT Region I - 2025

AWARDS & RECOGNITION

Pin Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2025)

Leadership Awardee – The Golden Flame, PHINMA University of Pangasinan (2025)

PEN Values Awardee: Commitment – The Golden Flame, PHINMA University of Pangasinan (2025)

Hard Hat Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2024)

Leadership Awardee – PHINMA UPang Student Leaders' Synergy and Excellence (PULSE) Awards (2024)



PHINMA UNIVERSITY OF PANGASINAN

APPENDIX B

CURRICULUM VITAE

MA. RHOABEL SANCHEZ

Lucban St, San Carlos City, Pangasinan, Philippines

math.sanchez.up@phinmaed.com

www.linkedin.com/in/mriarhoabel



EDUCATION

PHINMA UNIVERSITY OF PANGASINAN

College – Bachelor of Science in Computer Engineering

2022 – Present

CERTIFICATION

Safety Officer 2

IBESO – November 24, 2024

WORK EXPERIENCE

On-The-Job Trainee

Layad Circuits Electronics Engineering Supplies and Services

Baguio City, Benguet, Philippines

May 5, 2025 – June 20, 2025

- Assisted in planning and executing technical projects from concept to final delivery,
- Broke down project scope into actionable tasks and monitored weekly progress.
- Supported quality assurance through systematic testing and issue resolution.
- Managed project documentation, reports, and compliance requirements

SKILLS AND INTERESTS

Technical Skills

Programming & Development: Python, C++, Java, HTML, CSS, MySQL

Embedded Systems & Hardware: Basic Electronics, Breadboarding, ESP32 & Arduino Prototyping

Operating Systems & Tools: Windows, Git, Github, Microsoft Office, Basic IT Systems, Software Troubleshooting

Soft Skills

- Leadership and Team Coordination
- Attention to Detail
- Adaptability and Willingness to Learn
- Problem-Solving and Critical Thinking
- Time Management and Task Prioritization
- Professionalism and Work Ethic



PHINMA UNIVERSITY OF PANGASINAN

- Decision-Making Ability
- Organizational Skills

Languages: English (Fluent), Filipino (Fluent)

Interests: Data Engineering, Software Development, Programming, Leadership, Prototyping

LEADERSHIP

President

August 2025

– Present

Institute of Computer Engineers of the Philippines – Student Edition Region 1 Chapter

President

June 2024 – May 2025

Institute of Computer Engineers of the Philippines – Student Edition, PHINMA University of Pangasinan

2nd Year Representative

June 2023 – May 2024

Institute of Computer Engineers of the Philippines – Student Edition, PHINMA University of Pangasinan

SEMINARS, TRAININGS, AND ACTIVITIES

Data Analytics Essentials

Cisco Networking Academy – December 5, 2025

Internet of Things Conference 2025

Packetworx – September 10, 2025

Hack The Chain Ep25.2

Makerspace InnovHub – July 9-13, 2025

ICpEP 7th Regional Convention and CPE Challenge

Institute of Computer Engineers of the Philippines Region 1 Chapter – April 28, 2025

Arweave PH – Balangay Nationwide Roadshow

Makerspace InnovHub – January 26, 2025

Artificial Intelligence in Data Analytics

Department of Information and Communications Technology – Marinduque – January 26, 2025 (Webinar)

CpE InfoCruise 2025: Beyond Algorithms – Exploring the Overlooked Path of Cybersecurity

PHINMA UPang Institute of Computer Engineers of the Philippines – Student Edition – January 15, 2025

AWARDS & RECOGNITION

Pin Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2025)

Leadership Awardee – The Golden Flame, PHINMA University of Pangasinan (2025)

PEN Values Awardee: Competence – The Golden Flame, PHINMA University of Pangasinan (2025)

Leadership Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2024)

Hard Hat Awardee – CEA Success Ladder, PHINMA University of Pangasinan (2024)



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APPENDIX C

CODE

Machine Learning Pseudocode

START

1. Load system configuration

- Load camera source
- Load parking slot and zone coordinates
- Load vehicle detection model
- Initialize tracking module

2. Start camera capture thread

LOOP continuously:

 Read frame from IP camera

 IF frame read fails:

 Reconnect camera stream

3. Process each frame

- Apply optional cropping
- Detect vehicles using YOLO model
- Filter detections based on class, confidence, and geometry
- Track detected vehicles using ByteTrack

4. Determine slot occupancy



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FOR each detected vehicle:

 Compare vehicle bounding box with each parking slot polygon

 IF overlap exceeds threshold:

 Mark slot as occupied

5. Apply temporal smoothing

- Store occupancy results for several frames
- Confirm slot as occupied only if repeated consistently

6. Generate parking summary

- Count total slots
- Count occupied slots
- Count available slots
- Count blocked slots

7. Save and transmit results

- Write merged slot status to JSON file
- Send slot status to backend API

8. Display annotated output

- Draw bounding boxes
- Draw slot outlines
- Show slot labels: Available, Occupied

REPEAT until system stops

END



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Core Machine Learning Functions

```
# =====
# PYTHON: VEHICLE DETECTION
# =====

def detect_vehicles(frame, model, model_ref, detection_device, tracker=None):
    detections = []

    results = model.predict(
        source=[frame],
        conf=CONF_THRES_MAIN,
        device=detection_device,
        save=False,
        verbose=False,
    )

    boxes = results[0].boxes
    for box in boxes:
        conf = float(box.conf.cpu().item())
        bb = box.xyxy.detach().cpu().numpy()[0]

        if conf >= CONF_THRES_MAIN:
            detections.append({
                "conf": conf,
                "xyxy": (bb[0], bb[1], bb[2], bb[3]),
                "foot": bottom_center(bb),
            })

    return detections

# =====
# PYTHON: SLOT OCCUPANCY CHECK
# =====

def compute_slot_scores(state, detections):
    scores = defaultdict(float)
```



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```
for det in detections:
    conf = float(det["conf"])
    for s in state["slots"]:
        overlap = box_overlap_ratio_polygon(det["xyxy"], s["_contour"])
        if overlap >= BOX_OVERLAP_MIN and conf > scores[s["id"]]:
            scores[s["id"]] = conf

return scores

# =====
# PYTHON: OCCUPANCY SMOOTHING
# =====

def merge_slot_scores(global_slot_ids, slot_scores_by_cam, cam_weights, history):
    accum = {sid: [0.0, 0.0] for sid in global_slot_ids}

    for cam_idx, scores in enumerate(slot_scores_by_cam):
        weight = cam_weights[cam_idx]
        for sid, score in scores.items():
            accum[sid][0] += score * weight
            accum[sid][1] += weight

    for sid in global_slot_ids:
        total, wsum = accum[sid]
        merged_score = total / wsum if wsum > 0 else 0.0
        history[sid].append(merged_score)

    slot_status = {}
    for sid in global_slot_ids:
        hist = history[sid]
        avg_score = sum(hist) / len(hist) if hist else 0.0
        occupied = avg_score >= OCCUPIED_SCORE_THRESHOLD
        slot_status[sid] = {
            "occupied": occupied,
            "score": round(avg_score, 4),
```



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```
}

return slot_status

# =====
# PYTHON: BLOCKING DETECTION
# =====
def compute_blocking_scores(state, detections, global_slot_status=None):
    scores = defaultdict(float)
    blockers = {}

    for s in state["slots"]:
        sid = s["id"]
        block_contour = s.get("_block_contour")
        if block_contour is None:
            continue

        best_score = 0.0
        best_meta = None

        for det in detections:
            overlap = box_overlap_ratio_polygon(det["xyxy"], block_contour)
            if overlap >= BLOCKING_ZONE_OVERLAP_MIN:
                conf = float(det.get("conf", 1.0))
                score = overlap * conf
                if score > best_score:
                    best_score = score
                    best_meta = {
                        "track_id": det.get("track_id"),
                        "cls": det.get("cls"),
                        "score": round(score, 4),
                    }

        if best_meta is not None:
            scores[sid] = best_score
```



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```
blockers[sid] = best_meta
```

```
return scores, blockers
```

```
# =====
# PYTHON: CAMERA AUTO-RECONNECT
# =====
def start_capture_thread(state, grab_skip=0):
    def _reader():
        cap = state["cap"]

        while True:
            if state.get("stop_capture"):
                break

            ret, frame = cap.read()

            if not ret or frame is None:
                cap.release()
                time.sleep(RECONNECT_DELAY_SECONDS)
                cap = open_video_capture(
                    state["source"],
                    state.get("is_rtsp", False),
                    frame_size=state.get("frame_size"),
                    fps=state.get("fps"),
                    use_ffmpeg_nvdec=state.get("use_ffmpeg_nvdec"),
                )
                state["cap"] = cap
                continue

            with state["frame_lock"]:
                state["latest_ret"] = ret
                state["latest_frame"] = frame
                state["latest_ts"] = time.time()
```



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```
# =====  
# PYTHON: WRITE STATUS TO JSON  
# =====  
def write_merged_status(path, slot_status, summary):  
    payload = {  
        "timestamp": datetime.now().isoformat(timespec="seconds"),  
        "summary": summary,  
        "slots": slot_status,  
    }  
  
    with open(path, "w", encoding="utf-8") as f:  
        json.dump(payload, f, indent=2)
```

Backend Slot Detection and Status Management

```
<?php  
  
// =====  
// PHP: SINGLE DETECTION PROCESS  
// =====  
public function store(Request $request, ParkingSessionService $parkingSessionService)  
{  
    $validated = $request->validate([  
        'camera_id' => 'nullable|integer',  
        'slot_id'   => 'nullable',  
        'occupied'  => 'nullable|boolean',  
        'blocked'   => 'nullable|boolean',  
        'confidence' => 'nullable|numeric',  
    ]);  
  
    $slotId = $validated['slot_id'] ?? null;  
    $statusId = $this->resolveSlotStatusId($validated);  
    $confidence = $validated['confidence'] ?? 0.0;  
    $detectedAt = now();
```



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```
$detection = MISlotDetection::create([
    'camera_id'    => $validated['camera_id'],
    'slot_id'      => $slotId,
    'slot_status_id' => $statusId,
    'confidence'   => $confidence,
    'detected_at'  => $detectedAt,
]);

if ($slotId) {
    $slot = Slot::find($slotId);

    if ($this->shouldApplyMITransition($slot, $statusId, $detectedAt)) {
        $this->applyMITransition($slot, $statusId, $detectedAt);
    }
}

return response()->json([
    'message' => 'Detection recorded',
    'data' => $detection
]);
}

// =====
// PHP: BATCH DETECTION PROCESS
// =====

public function storeBatch(Request $request, ParkingSessionService
$parkingSessionService)
{
    $items = $this->normalizeBatchItems($request);
    $rows = [];

    foreach ($items as $item) {
        $slotId = $item['slot_id'];
        $statusId = $this->resolveSlotStatusId($item);
```



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```
$rows[] = [
    'camera_id'    => $item['camera_id'],
    'slot_id'      => $slotId,
    'slot_status_id' => $statusId,
    'confidence'   => $item['confidence'] ?? 0.0,
    'detected_at'  => $item['detected_at'] ?? now(),
];
}

MISlotDetection::insert($rows);

foreach ($rows as $row) {
    $slot = Slot::find($row['slot_id']);

    if ($this->shouldApplyMITransition($slot, $row['slot_status_id'],
    $row['detected_at'])) {
        $this->applyMITransition($slot, $row['slot_status_id'], $row['detected_at']);
    }
}

return response()->json([
    'message' => 'Batch processed successfully',
    'count' => count($rows)
]);
}

// =====
// PHP: STATUS DECISION LOGIC
// =====

private function shouldApplyMITransition(Slot $slot, int $newStatusId, Carbon
$detectedAt): bool
{
    if ($slot->slot_status_id === $newStatusId) {
        return false;
    }
}
```



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```
}

if (!$this->hasConsecutiveDetections($slot->id, $newStatusId, 3)) {
    return false;
}

$lastUpdate = $slot->last_ml_update;
if ($lastUpdate && $detectedAt->diffInSeconds($lastUpdate) < 10) {
    return false;
}

return true;
}

// =====
// PHP: APPLY SLOT STATUS UPDATE
// =====
private function applyMlTransition(Slot $slot, int $newStatusId, Carbon $detectedAt):
void
{
    $slot->update([
        'slot_status_id' => $newStatusId,
        'last_ml_update' => $detectedAt,
    ]);
}

// =====
// PHP: CONSECUTIVE DETECTION CHECK
// =====
private function hasConsecutiveDetections(int $slotId, int $statusId, int
$requiredFrames): bool
{
    $detections = MISlotDetection::where('slot_id', $slotId)
        ->orderByDesc('detected_at')
```



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```
->limit($requiredFrames)
->get();

if ($detections->count() < $requiredFrames) {
    return false;
}

return $detections->every(function ($d) use ($statusId) {
    return $d->slot_status_id === $statusId;
});
}
?>
```

ML Car Detection and QR-Based Vehicle Identification

```
<?php

// =====
// ML CAR DETECTION (MAIN FLOW)
// =====

public function store(Request $request)
{
    // 1. Validate incoming data
    $validated = $request->validate([
        'camera_id' => 'required|integer',
        'qr_value' => 'nullable|string',
        'slot_id' => 'nullable',
        'confidence' => 'required|numeric|min:0|max:1',
    ]);

    // 2. Process detection
    return $this->recordDetection($validated);
}
```



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```
// =====
// CORE PROCESSING FUNCTION
// =====
private function recordDetection(array $data)
{
    // 3. Resolve QR → User/Vehicle
    $qr = UserQrCode::where('qr_value', $data['qr_value'])
        ->where('qr_status_id', 1)
        ->first();

    if (!$qr) {
        return response()->json([
            'message' => 'Invalid or expired QR'
        ], 422);
    }

    // 4. Extract vehicle info
    $vehicleId = $qr->vehicle_id ?? null;

    // 5. Save detection
    $detection = MlCarDetection::create([
        'camera_id' => $data['camera_id'],
        'slot_id' => $data['slot_id'] ?? null,
        'user_qr_code_id' => $qr->id,
        'vehicle_id' => $vehicleId,
        'confidence' => $data['confidence'],
        'detected_at' => now(),
    ]);

    // 6. Trigger parking session logic
    app(ParkingSessionService::class)
        ->ingestDetection($detection);

    // 7. Return response
```



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```
return response()->json([
    'message' => 'Car detection recorded successfully',
    'data'    => $detection
], 201);
}
```

```
// =====
// OPTIONAL: QR SCAN → ML QUEUE
// =====
public function scanToMl(Request $request)
{
    $validated = $request->validate([
        'qr_value' => 'required|string',
        'camera_id' => 'nullable|integer',
    ]);

    // Queue QR for ML processing
    $queueItem = PendingQrScan::create([
        'qr_text' => $validated['qr_value'],
        'payload' => $validated,
    ]);

    // Dispatch job to ML system
    SendPendingQrToMl::dispatch($queueItem->id);

    return response()->json([
        'message' => 'QR sent to ML',
        'queue_id' => $queueItem->id
    ]);
}
```